

Data Governance Onboarding - INVX

Data Quality Management

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DATA TO COVER THE 5 KEY COMPONENTS OF DATA GOVERNANCE



Data Governance Organization & Policy

- Data Organization that will drive the realization and sustainment of data capabilities
- Data governance policy, standard & processes in line with regulatory compliance



Data Sharing

- Data Sharing refers to data provisioning and data access:
- Data provisioning is based on Exclusion List which includes:
 - ✓ SCBx and Monolines business interest
 - ✓ Regulatory requirements (PDPA, BOT...)
- Data access is based on user roles, purpose of use and customer consent



Metadata Management

- Metadata is information about data e.g., definition, name, format, storage location, quality...
- Metadata includes two components:
 - ✓ Business
 - ✓ Technical



Data Quality

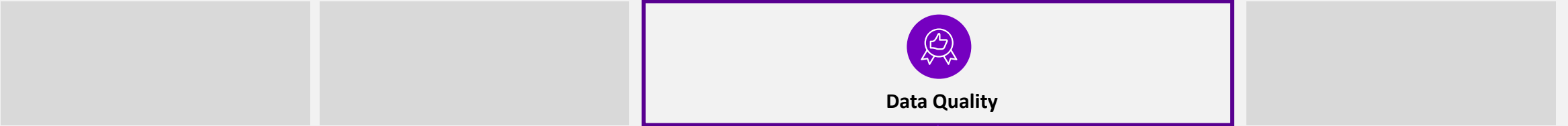
- Data Quality is the ability of data to satisfy the stated business, regulatory, system, and technical requirements
- Data Quality measured by various aspects:
 - ✓ Completeness
 - ✓ Conformity
 - ✓ Integrity
 - ✓ Uniqueness
 - ✓ Consistency
 - ✓ Accuracy
 - ✓ Timeliness



Data Lineage

- Data Lineage is the process of analyzing data as it flows from data sources to consumption enabling audit and data traceability

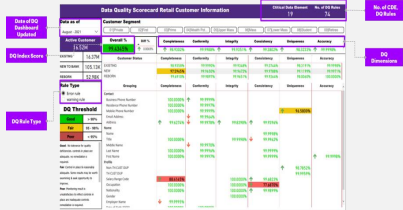
AS A PILLAR OF DATA GOVERNANCE, ROBUST DATA QUALITY MANAGEMENT NEEDS TO BE RIGHT WITH BOTH STANDARDS & PROCESSES



Data Quality Standard

Agreed definition, format & representation to unify approach to select and process data, and examine data quality

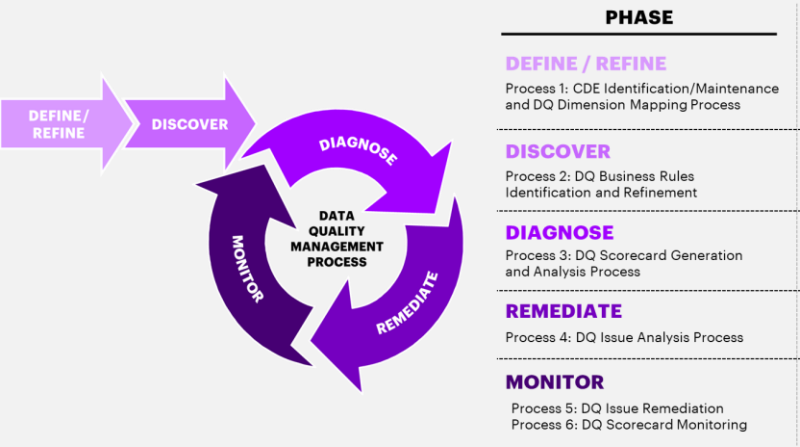
Dimension	Description	Example
1. Completeness	Data must be not missing or unusable	"Customer Name" must have value
2. Conformity	Data must be stored in required format	"RN_ID" must contain 10 digits and start with "0001", "0002", "0003"
3. Integrity	Data must be referenced to a valid set of predefined 'allowable' values	"Title" must have value according to reference lists only
4. Uniqueness	Data must not be duplicated and must be the only one of its kind	"Employee ID" cannot be duplicated or mapped to more than 1 name
5. Consistency	Data must not give conflicting information	"Title" and "Gender" must be consistent
6. Accuracy	Data must be correct and not out of date	"Birth Date" value cannot be defined as a future date compared with data captured in Data Lake and calendar data
7. Timeliness	Data must be available when it is expected to be used	Data table must be available and number of records acceptable



No	Data Attribute Name	System Name	Data Set Name	Column Name	Data Quality Dimension	Rule ID	Rule Status	Business Rules	Rule Type	Conditions
1	Employee ID	B2P	Emp	Emp_ID	Completeness	Emp_ID_1	New	Employee ID must have value for 5 digits	Error	Employee ID value must contain 5 digits only Pass: Contain 5 digits of number Fail: Blank or NULL
2	Employee ID	B2P	Emp	Emp_ID	Conformity	Emp_ID_2	Update	Employee ID must not contain only number	Error	Employee ID value must contain only number Pass: Contain number Fail: Contain letter, space or special character

Data Quality Process

Standardized operations for data quality governance across Group Data Zone and Monoline Data Zone



STRONG DATA QUALITY MANAGEMENT IS A **KEY DRIVER** IN UNLOCKING THE FULL POTENTIAL OF DATA

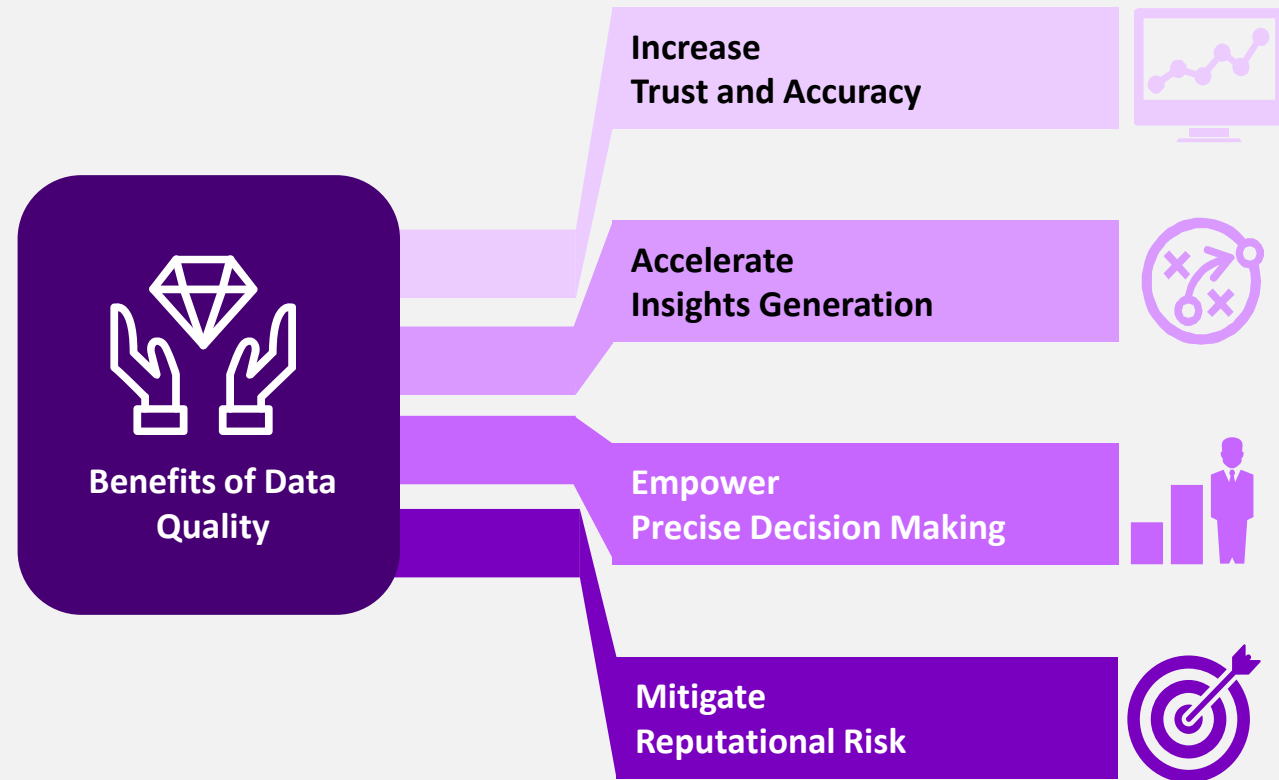
Need for Data Quality

Data is an integral part of business which needs to be monitored, measured and analyzed. The **higher the level of data quality**, the more **relevant and effective** the measures, analysis results and resultant actions.

Data Quality Management:

- Focuses on measuring and monitoring data quality levels, analyzing impacts of issues, on resolving root causes and fixing data
- Provides consistent data quality processes across organization

Benefits of Strong Data Quality Management



KEY PRINCIPLES OF DATA QUALITY

1

**Proactive
Quality Maintenance**

Data Quality is an **ongoing initiative** that focuses on **solving root causes of Data Quality challenges**

2

**Accountable
Quality Management**

Identify and appoint roles with key responsibilities of upholding high Data Quality standards in the organization

3

**Clear Processes
for Governance**

Data Quality in the organization systems should be **measured, reported, documented and monitored** through **robust governance** processes

4

**Measurable and Trackable
Outcomes**

Data Quality must be **quantifiable** via **metrics** to track the progress of data improvement efforts

5

High Transparency

Data Quality metrics/indices should **provide transparency** on current status of data quality to **increase stakeholders' confidence** in operational, analytical and enterprise-wide reported data

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IS THERE ANY KEY PAIN POINT OR BLOCKER ? THAT CAN ADD VALUE TO FURTHER STRENGTHEN AND STANDARDIZE DATA QUALITY



As-is Key Findings



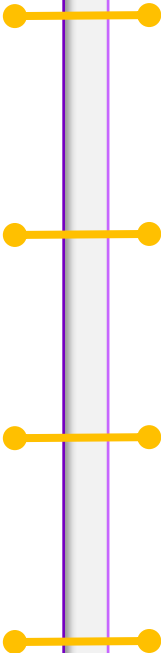
Opportunities



Data Quality

Example :

- *the standards and processes not yet in place*
- *there are no specific roles of Data Quality Officers or Data Stewards appointed*
- *Monolines able to leverage existing tool for data quality management or not*
- *Data Quality tracking is in place for not under DGO*



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CRITICAL DATA ELEMENT IS THE BUILDING BLOCK FOR DATA QUALITY MANAGEMENT

Critical Data Elements

Critical Data Elements (CDEs) are the key data used by DataX and Monolines for their **operational decision making**, **internal or external reporting** or to meet **regulatory requirements**.

The CDEs can be defined as below*:



Regulation

- Data that is mandated for **regulatory requirement**
- Data that is used for the **business policy**



Report

- Data that is used for **both internal and external report**
- Data that is used for **management decision making**



Business Operations

- Data that is used for **business operation**, esp. activities impacting customer satisfaction
- Data that is used for **operational decision making**, e.g., financial and customer data for business goal setting

* Note: These criteria, as adapted from SCB’s existing DQ guidelines, apply across DataX and monolines, while identification of CDEs are based on use cases

Data Subject Area	ID	Data Element
Customer Data	1	Customer_ID
	2	Customer_DOB
	3	Customer_Address
	4	Customer_PhoneNumber
	5	Customer_email
	6	Customer_CountryOfResidence
	7	Customer_LoyaltyValue
	8	Customer_Revenue
	9	Customer_Password
	10	Customer_Website_ID
	11	Customer_Registered_Date
	12	Customer_Approved_Date
	13	Customer_Updated_Date
	14	Customer_Updated_Timestamp

Critical Data Element

Non-Critical Data Element

(OPTIONAL) WE RECOMMEND **CDE PRIORITISATION** APPROACH BASED ON THE SIGNIFICANCE OF BUSINESS IMPACT

Critical Data Elements (CDEs) are the key data used by DataX and Monolines for their **operational decision making, internal or external reporting** or to meet **regulatory requirements**.

The CDEs can be defined as below*:

**Regulation**

- Data that is mandated for **regulatory requirement** ★
- Data that is used for the **business policy**

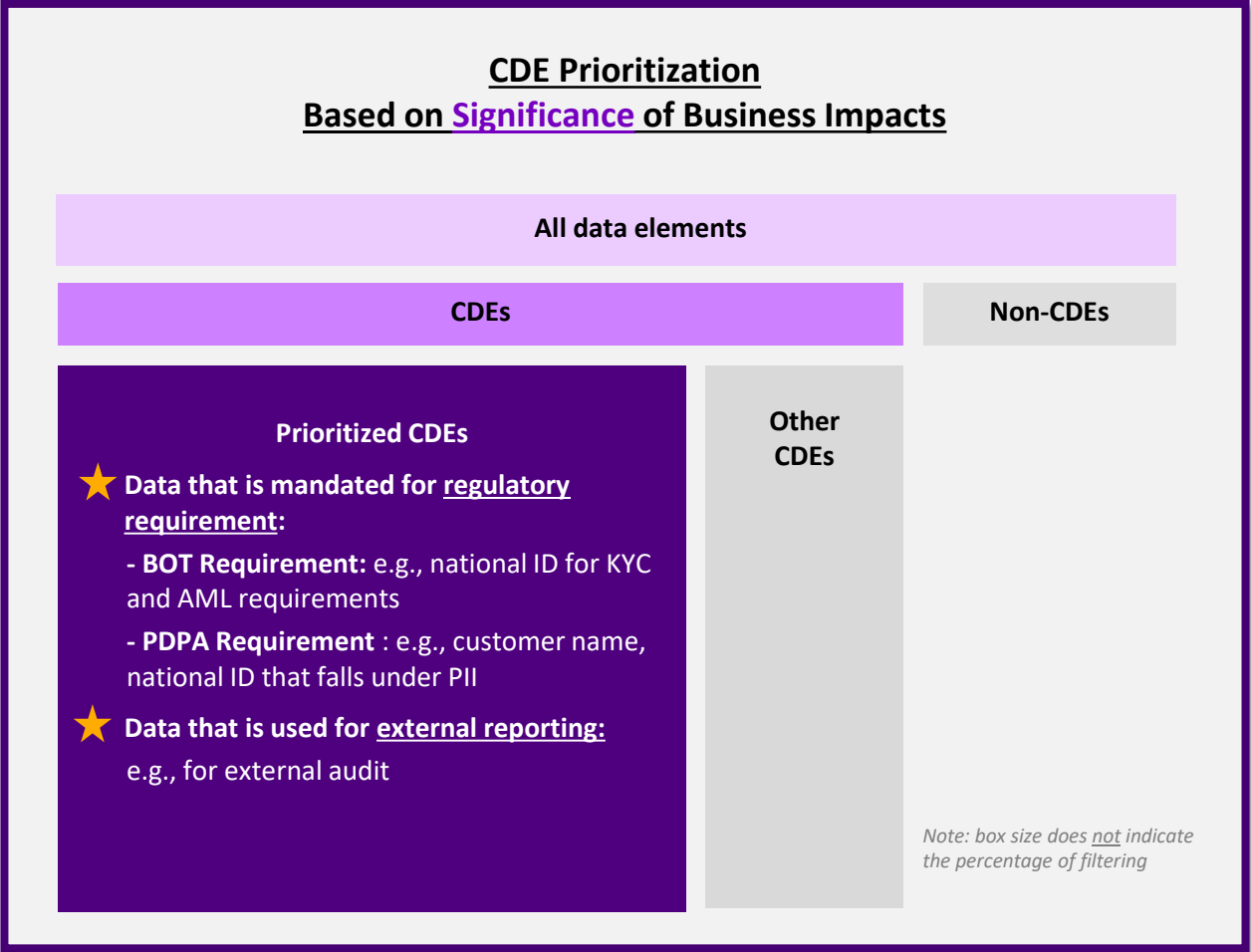
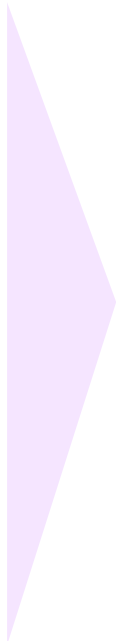
**Report**

- Data that is used for **both internal and external report** ★
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**Business Operations**

- Data that is used for **business operation**, esp. activities impacting customer satisfaction
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* Note: These criteria, as adapted from SCB's existing DQ guidelines, apply across DataX and monolines, while identification of CDEs are based on use cases



DATA QUALITY DIMENSION DEFINES THE KEY METRICS TO GUIDE ASSESSMENT

Dimension	Description	Example			
1. Completeness	Data must be not missing or unusable	“Customer Name” must have value	Customer Name		
			Somchai		
2. Conformity	Data must be stored in required format	“Date of Birth” must match YYYY-MM-DD format	Date of Birth		
			1970-12-25		
			3000-29-02		
3. Integrity	Data must be referenced to a valid set of predefined ‘allowable’ values	“Title” must have value according to reference lists only	Title Reference	TitleCustomer Name	
			Miss	Miss	Somying
			Mister	Khun	Somchai
4. Uniqueness	Data must not be duplicated and must be the only one of its kind	“Employee ID” cannot be duplicated or mapped to more than 1 name	Employee IDEmployee Name		
			12345	Somchai	
			12345	Somying	
5. Consistency	Data must not give conflicting information	“Title” and “Gender” must be consistent	TitleGender		
			Mister	Male	
			Miss	Male	
6. Accuracy	Data must be correct and not out of date	“Date of Birth” value cannot be defined as a future date compared with data captured in Data Lake and calendar date	Date of Birth		
			1970-12-25		
			3000-29-02		
7. Timeliness	Data must be available when it is expected to be used	Dataset “Fes_txn_dly“ must be available and number of records are acceptable	Data Set	dl_data_dt	Records
			Fes_txn_dly	2022-05-30	20,000,000
			Fes_txn_dly	2022-05-31	0



DATA QUALITY BUSINESS RULE TEMPLATE IS TO STANDARDIZE THE APPROACH TO SETTING DATA QUALITY BUSINESS RULES (1/2)

1	2	3	4	5	6	7	8	9	10	11
No	Data Element Name	System Name	Data Set Name	Column Name	Data Quality Dimension	Rule ID	Rule Status	Business Rule Description	Rule Type	Business Condition
1	Employee ID	B2P	Emp	Emp_ID	Completeness	Emp_ID_1	New	Employee ID must have value for 5 digits	Error	Employee ID value must contain 5 digits only Pass: Contain 5 digits of number Fail: Blank or NULL
2	Employee ID	B2P	Emp	Emp_ID	Conformity	Emp_ID_2	Update	Employee ID must not contain only number	Error	Employee ID value must contain only number Pass: Contain number Fail: Contain letter, space or special character

No	Fields	Explanation
1	No	Running number
2	Data Element Name	Name of the data element
3	System Name	This is a system that contains the data set name and column name. This can include the originating system as well as other systems that contain the data element in its data set. This system may contain relational or non-relational data sets.
4	Data Set Name	Data set is a collection of information within a system. This data set can be relational (e.g., table) or non-relational (e.g., CSV, .DAT)
5	Column Name	Name of the data column in the data set. In case of non-relational data sets, this column might not be relevant.
6	Data Quality Dimension	Type of defined data quality dimension
7	Rule ID*	Data quality business rule ID
8	Rule Status**	Status of business rules: new, update, delete, no change
9	Business Rule Description	Data quality business rule description for reporting purpose
10	Rule Type	Data quality rule type: error, warning
11	Business Condition	Business condition on how to measure the data quality business rule

*Rule ID naming convention is open to the design from developing team
** The Rule Status will be defined according to the version control list of Rule Status

DATA QUALITY BUSINESS RULE TEMPLATE IS TO STANDARDIZE THE APPROACH TO SETTING DATA QUALITY BUSINESS RULES (2/2)

12	13	14	15	16	17	18	19
Updated Date	Updated By	Additional Remarks	Related Regulation	BOT Request	AML Request	Other Regulation (Specify)	Audit Finding Noted (Year-Team)
TBC	TBC	TBC	Yes	Yes	Yes	N/A	N/A
TBC	TBC	TBC	No	Yes	Yes	N/A	N/A

No	Fields	Explanation
12	Updated Date	The date when latest change is registered to data quality business rules, e.g. business rules or conditions have been changed
13	Updated By	Person who registered the latest changes on the data quality business rules
14	Additional Remarks	Description of the changes made in the data quality business rules
15	Related Regulation	To specify if there is any regulation related to the business rule or not: Yes/ No
16	BOT Request*	To specify if the business rule is required by BOT: Yes/ No
17	AML Request*	To specify if the business rule is required by AML: Yes/ No
18	Other Regulation (Specify)*	To specify if the business rule is required by any regulation besides BOT and AML
19	Audit Finding Noted (Year-team)	To specify the audit finding of the business rule

*The column names are subject to change depending on the focus area of regulation e.g., PDPA, PII

GLOBAL DATA QUALITY INDEX THRESHOLD SETS THE STANDARDS FOR DATA QUALITY CONTROL

Global Data Quality Index Threshold



Good

> 98%

- DQ fully meets expectations
- No tolerance for DQ issues to reach this threshold
- No further remediation plan required, DQ controls are adequate



Fair

95 – 98%

- DQ does not fully meet expectations
- Higher quality standards are required in this threshold
- Remediation plan may be required, controls should be reviewed to improve DQI



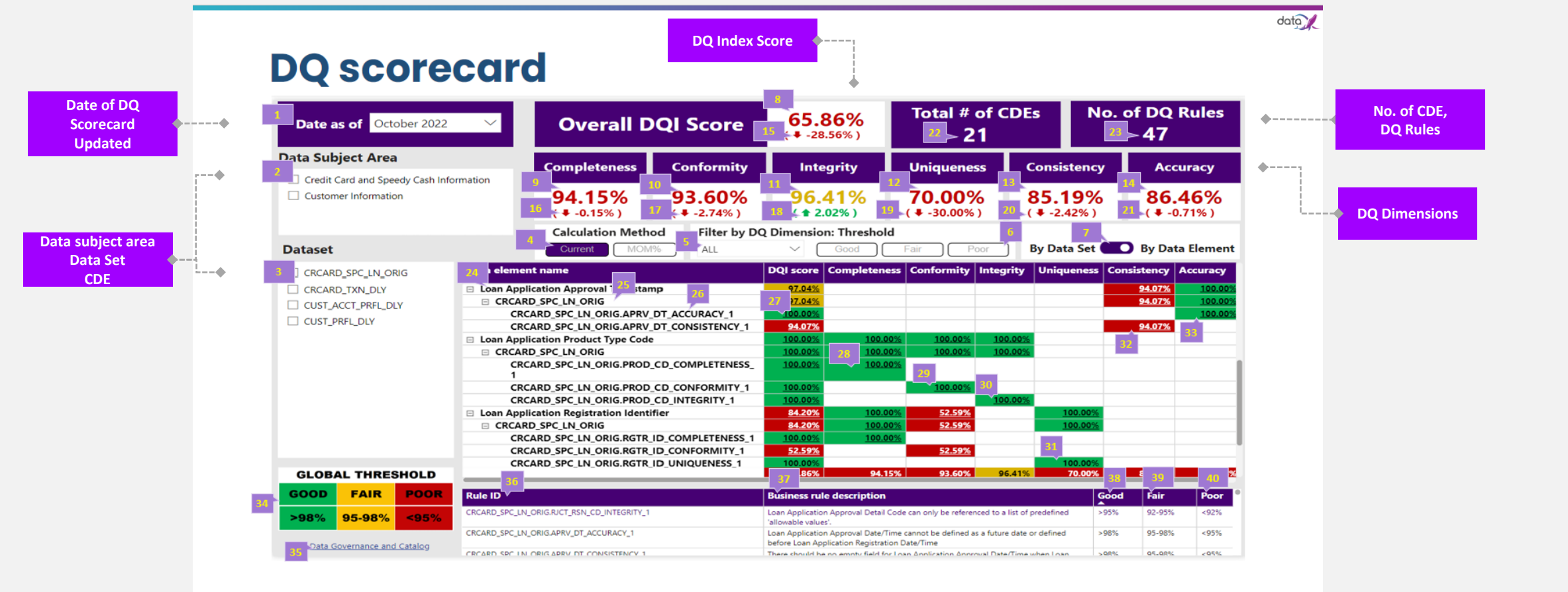
Poor

< 95%

- DQ does not meet expectations
- Strong remediation plan is required to improve DQI
- Controls need to be improved

DATA QUALITY SCORECARD* ALLOWS EASY VISUALIZATION, MEASUREMENT, AND MONITORING OF DATA QUALITY

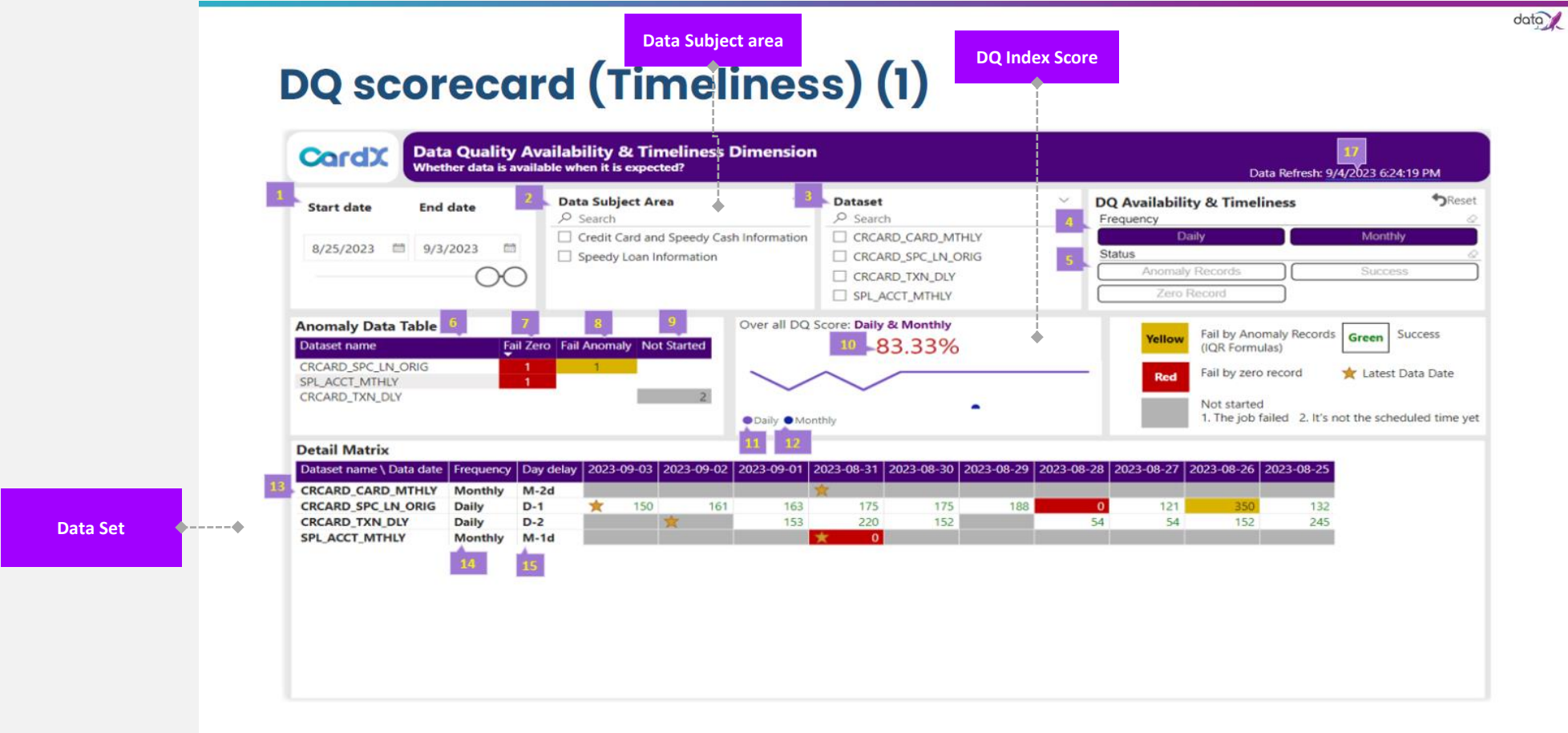
Users can easily evaluate the quality and trustworthiness of data against predetermined thresholds.



* This is a mock-up of Data Quality Scorecard. However, the actual standard scorecard layout and components is dependent on the selected Data Quality tool.

DATA QUALITY SCORECARD* ALLOWS EASY VISUALIZATION, MEASUREMENT, AND MONITORING OF DATA QUALITY

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* This is a mock-up of Data Quality Scorecard. However, the actual standard scorecard layout and components is dependent on the selected Data Quality tool.

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A **WELL-DEFINED SCOPE OF RESPONSIBILITY** FOR DATA_X & MONOLINES WILL SAFEGUARD DATA QUALITY MANAGEMENT ACROSS GDZ & MDZ

GROUP DATA ZONE (GDZ)

Data_X to manage the overall data quality in the Group Data Zone



- Define data quality standards & processes



- Assess data quality based on defined business rules and metrics



- Generate DQ scorecard to identify issues and guide investigation



- Coordinate root-cause analysis and remediation for prioritized DQ issues



- Monitor progress on remediation to ensure improvement of data quality



- Conduct regular review of DQ standards to guide refinement of DQ Mgmt.

MONOLINE DATA ZONE (MDZ)

Monolines to govern own data quality in the Monoline Data Zone
(in adherence to the standards & processes set by Data_X)

DATA QUALITY TEAM OF BOTH DATAX AND MONOLINES WILL WORK CLOSELY WITH CORE AND SUPPORTING STAKEHOLDERS



Roles Involved in DQ Management

(Applicable to DataX and Monolines)

Core Roles

- **Data Quality Team**
Define DQ standards, monitor and manage overall data quality
- **Data Owner**
Represent specific business or data subject area(s) and play DQ leadership role for own responsible data by enforcing DQ standards & processes
- **Business Data Steward**
Take charge of the execution of data quality management activities
- **Technical Data Steward**
Take charge of technical activities for DQ management

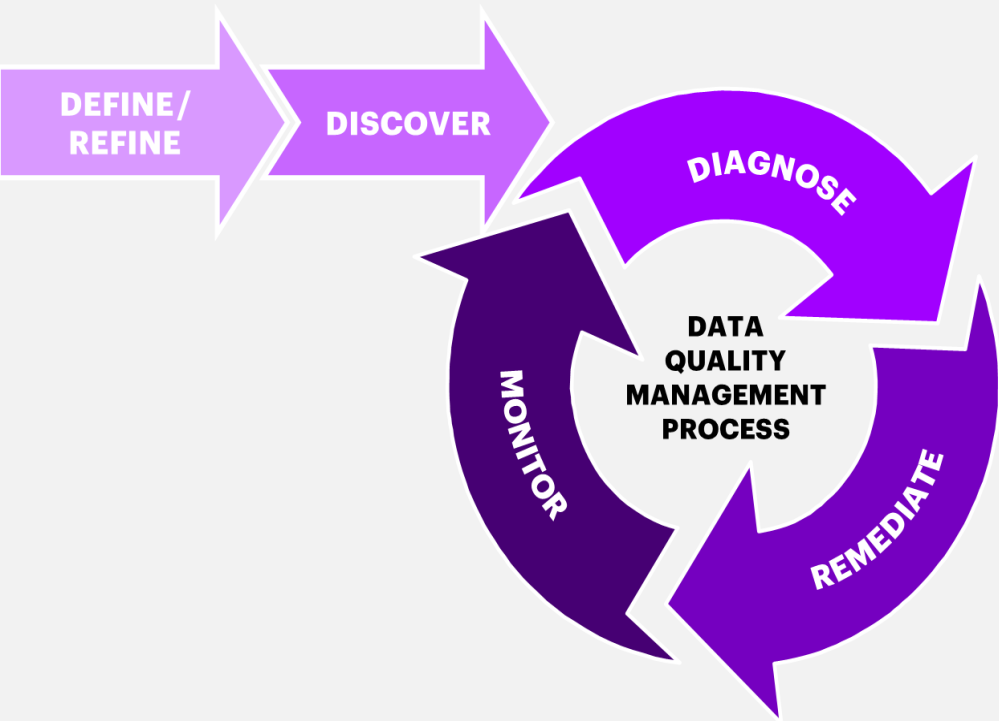
Supporting Roles

- **Heads of Data Governance**
Oversee DQ standards & processes, and be consulted on the overall data health* & DQ performance metrics
- **Metadata Management Officer**
Consulted during data quality profiling for clarification in GDZ/MDZ related to metadata and lineage
- **Data Protection Officer**
Work closely with DQ teams to ensure compliance with data protection, especially for issues impacting PII data
- **Other Relevant Stakeholder**
e.g., data producer(s) to support business data steward in root cause analysis

* DQ Working Group Forum also evaluates and reviews overall data quality index outcome

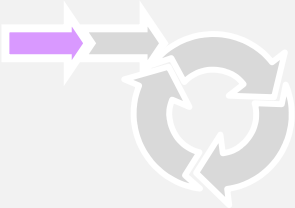


WE RECOMMEND A **PHASED APPROACH** TO OPERATIONALIZE AND CONTINUOUSLY STRENGTHEN DATA QUALITY MANAGEMENT



PHASE	DESCRIPTION
DEFINE / REFINE Process 1: CDE Identification/Maintenance and DQ Dimension Mapping Process	Define / refine foundational DQ components, incl. identification / maintenance of Critical Data Elements (CDEs) and mapping of DQ dimensions
DISCOVER Process 2: DQ Business Rules Identification and Refinement	Profile data, refine and confirm DQ business rules for implementation, and apply global DQ Index Threshold
DIAGNOSE Process 3: DQ Scorecard Generation and Analysis Process	Generate DQ scorecard and conduct high-level analysis to identify DQ issues
REMEDiate Process 4: DQ Issue Analysis Process	Prioritize issues and conduct root-cause and impact analysis for DQ issue
MONITOR Process 5: DQ Issue Remediation Process 6: DQ Scorecard Monitoring	Monitor progress on remediation, as well as continuously monitor DQ scorecard, to ensure improvement of data quality

DEFINE/REFINE PHASE IS A CRITICAL STEP TO SET UP AND STRENGTHEN KEY MECHANISM



?

What to be defined

2 DQ Foundational Components

1. Critical Data Elements (CDEs)

Critical Data Elements

Critical Data Elements (CDEs) are the data used by DataX and Monitors for their operational decision making, internal or external reporting or to meet regulatory requirements.

The CDEs can be defined as below:

Business Operations

Report

Regulation

• Data that is used for business operation

• Data that is used for operational decision making such as financial and customer data

• Data that is used for both internal and external report

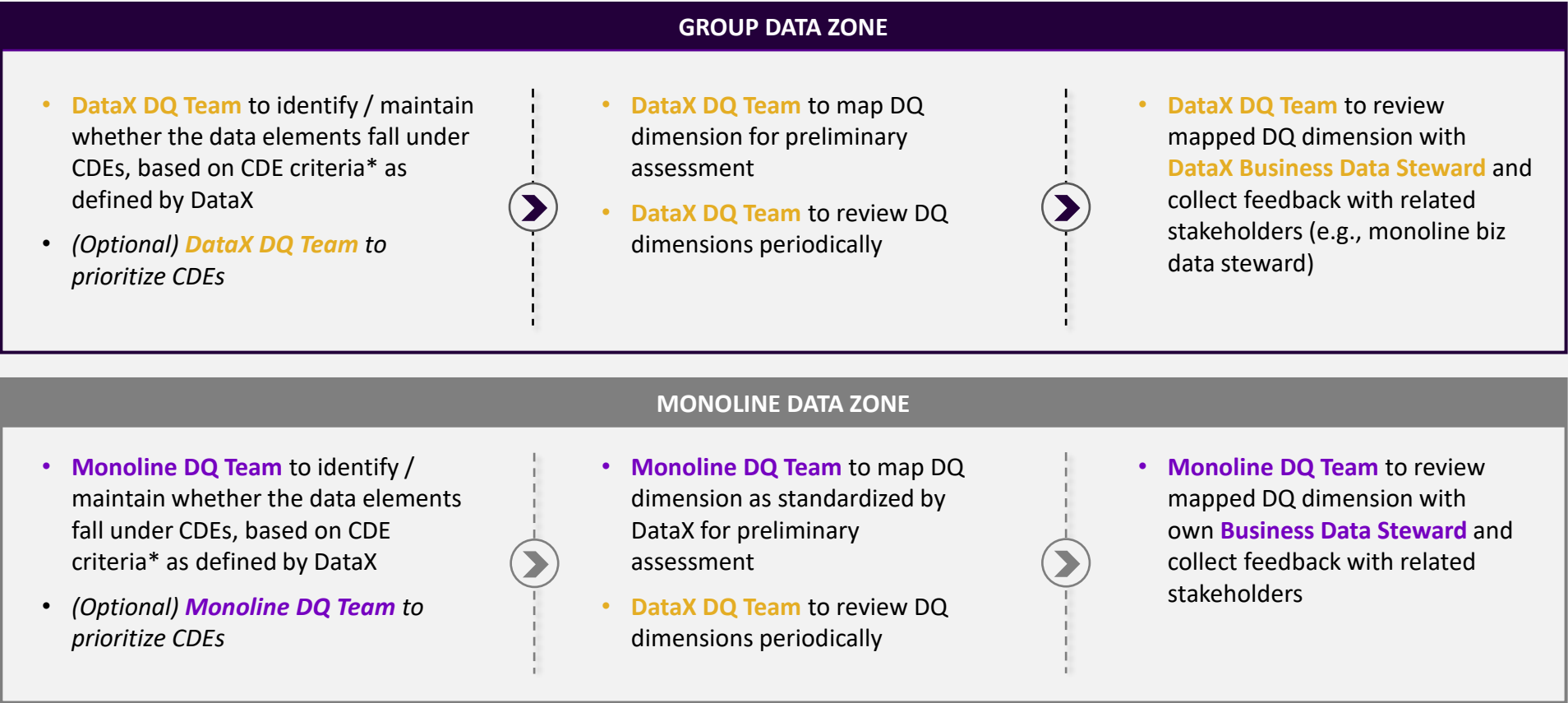
• Data that is used for regulatory requirement

• Data that is used for the business policy

Note: more detailed definition criteria can be defined as needed.

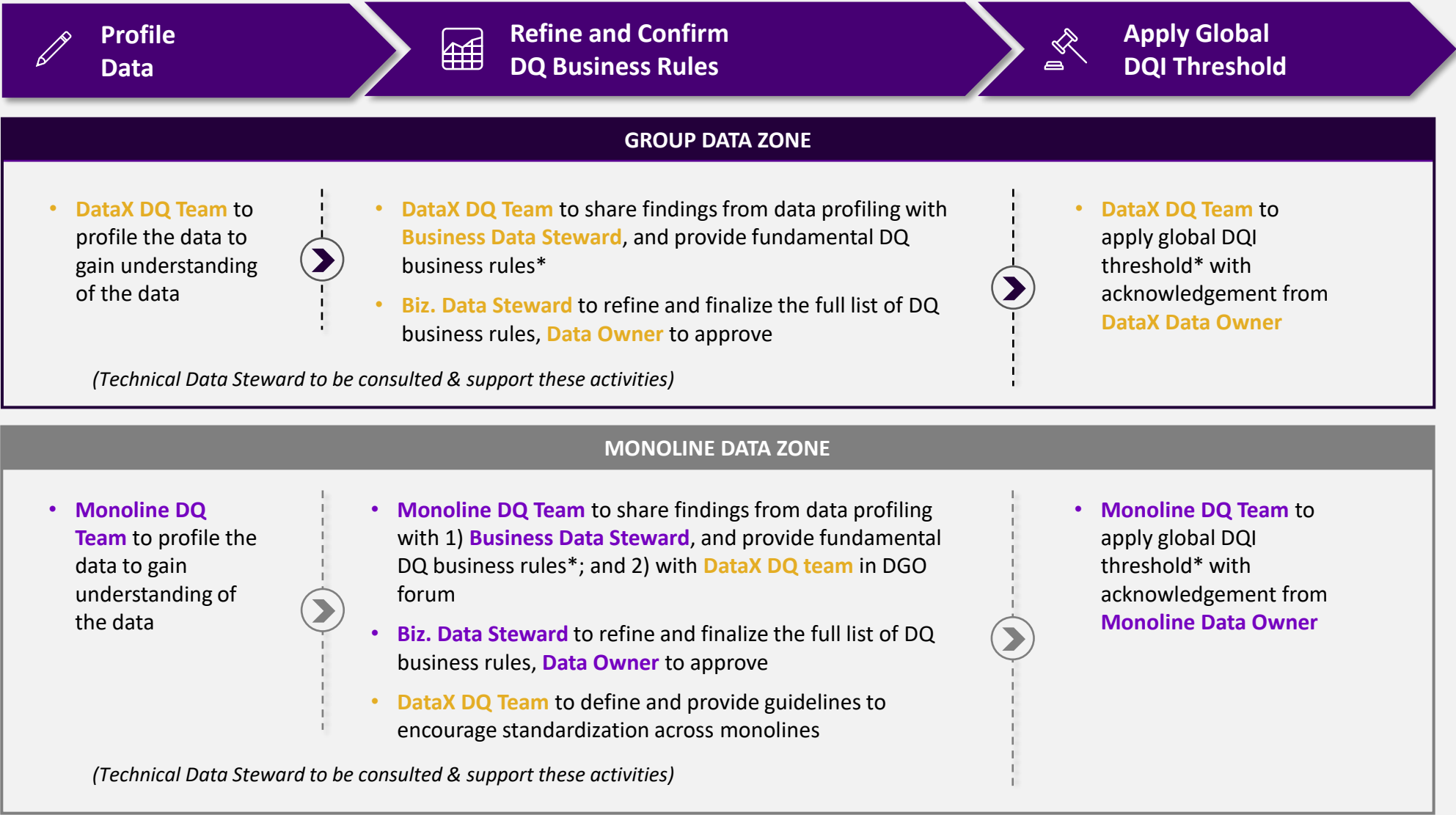
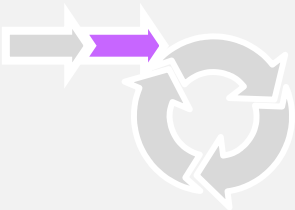
2. DQ Dimensions

Dimension	Description
1. Completeness	Data must be not missing or unusable
2. Conformity	Data must be stored in required format
3. Integrity	Data must be referenced to a valid set of predefined allowable values
4. Uniqueness	Data must not be duplicated and must be the only one of its kind
5. Consistency	Data must not give conflicting information
6. Accuracy	Data must be correct and not out of date
7. Timeliness	Data must be available when it is expected to be used



* CDE criteria (and prioritization criteria) will be regularly reviewed (based on agreed frequency) and continuously maintained

DISCOVER PHASE IS TO UNDERSTAND DATA AND SET RULES & TARGETS



* DQ Business Rules Template and Global DQI Threshold will be regularly reviewed (based on agreed frequency) and continuously maintained

DIAGNOSE PHASE GENERATES DQ SCORECARD TO GUIDE ANALYSIS



GROUP DATA ZONE

- **DataX DQ Team** to run DQ business rules and generate the standard DQ scorecard *(of derived data, aggregate data, and timeliness for monoline data)*
- **DataX DQ Team** to consolidate DQI performance results on overall data health

- **DataX DQ Team** to carry out analysis incl.:
 - Score analysis against pre-defined targets
 - Trend analysis
 - Anomaly and outlier identifications
- **DataX Data Owner** to adjust DQI Threshold to selected CDE (if required)

- **DataX DQ Team** to summarize and communicate findings to **DataX Data Owner**

MONOLINE DATA ZONE

- **Monoline DQ Team** to run own DQ business rules and generate own DQ scorecard (in a standardized template by DataX DQ team)
- **Monoline DQ Team** to consolidate own DQI performance results on overall data health

- **Monoline DQ Team** to carry out analysis incl.:
 - Score analysis against pre-defined targets
 - Trend analysis
 - Anomaly and outlier identifications
- **Monoline Data Owner** to adjust DQI Threshold to selected CDE (if required)

- **Monoline DQ Team** to summarize and communicate findings to own **Monoline Data Owner**

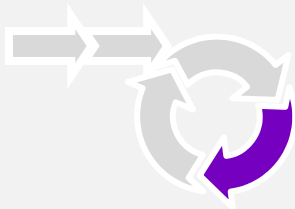
? Timeliness for Monoline Data

When provisioning data from MDZ to GDZ, DataX must ensure that the real-time and up-to-date data is timely provisioned (i.e., low/no latency)

```
graph LR; MDZ[MDZ Monoline Data] --> GDZ[GDZ Monoline Data];
```

* DQ Scorecard Components will be regularly reviewed (based on agreed frequency) and continuously maintained

REMEDiate PHASE IDENTIFIES THE CRITICAL DATA QUALITY ISSUES

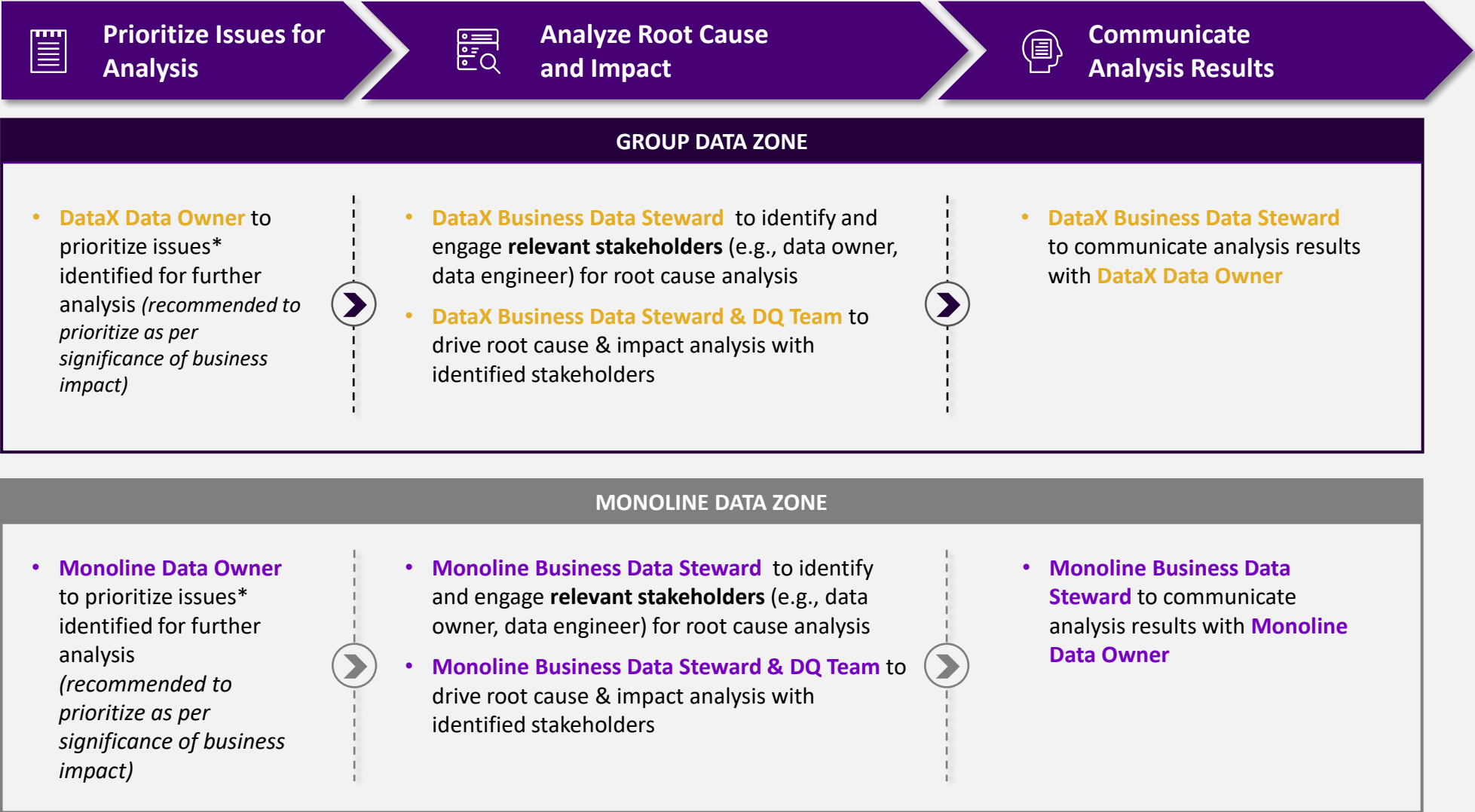


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Recap on How to Prioritize

We recommend prioritization based on significance of business impact

- For regulatory requirement:
 - BOT Requirement: e.g., national ID for KYC and AML
 - PDPA Requirement: e.g., PII
- For external reporting: e.g., customer's credit payment history to credit bureau



MONITOR PHASE IS TO ENSURE THE IMPROVEMENT OF DATA QUALITY



Remediation

Corrective Action

To specifically fix identified DQ issues in the system (short term)

Preventive Action

To systematically avoid (re-) occurrence of the problem (long-term)

GROUP DATA ZONE		
• DataX Business Data Steward to plan for DQ corrective action plan and update progress • DataX Data Owner to approve DQ corrective action plan • DataX DQ Team to suggest preventive actions based on remediation plan		• DataX Business Data Steward to identify and engage relevant stakeholders to execute both corrective and preventive actions for remediation • DataX DQ Team to monitor progress of execution of remediation plan
• DataX DQ Team to monitor DQ scorecard periodically based on agreed metrics • DataX DQ Team to communicate to relevant governance forums based on agreed frequency		

MONOLINE DATA ZONE		
• Monoline Business Data Steward to plan for DQ corrective action plan and update progress • Monoline Data Owner to approve DQ corrective action plan • Monoline DQ Team to suggest preventive actions based on remediation plan		• Monoline Business Data Steward to identify and engage relevant stakeholders to execute both corrective and preventive actions for remediation • Monoline DQ Team to monitor progress of execution of remediation plan
• Monoline DQ Team to monitor DQ scorecard periodically based on agreed metrics • Monoline DQ Team to communicate to relevant governance forums based on agreed frequency		

CORRECTIVE AND PREVENTIVE ACTIONS MUST BE PERFORMED TO CONTINUOUSLY MAINTAIN DATA IN GOOD QUALITY



Corrective Action

Actions to fix existing data quality by correcting the data in the system



Preventive Action

Actions to avoid the problem from occurring or re-occurring through improvement at the data entry point

Methods for Corrective Action:

Reactive	Proactive
One-Time Data Cleansing Activity	Scrub Campaign on Target Sources

Methods for Preventive Action:

Comms	Manpower	Method	Material	Machine
- Policies and higher-level decision making - Data Privacy and Protection Policy - Data quality awareness and data-driven culture - Training for data producers - Accessible documentation	- Knowledge Worker - Data Producers - Developers - Data analyst - Business Analyst - System Analyst	- Business Processes - Documented Procedures - Data analysis method - Application development method	- Standards and documentation - Input data (e.g., field forms, entry structure) - Materials used for development (e.g., forms, DQ reports, procedures)	- Modeling and development tools - Technical Transformation Rules - Applications - Databases - System software & Hardware

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Appendix A: Glossary of Terms

GLOSSARY OF TERMS (1/2) (BY ALPHABETIC ORDER)

Term	Definition
Aggregate Data	Data from the monolines or group data zone which is aggregated into groups or categories instead of being maintained at the transactional level (e.g., total purchases in last 6 months)
Corrective Action	A short-term action to specifically fix certain data governance issue(s)
Critical Data Element (CDE)	A key data element that is important to the success of a key business process
Curated Data	Data which has been standardized across the various sources in order to have a common definition and understanding of the same data regardless of source
Data Attribute	The name of the data attributes the same term as the data element.
Data Consumer	Any person who consumes / utilizes the data available on GDZ. Example - Business User, Business Data Steward, Data Engineer, Data Scientist
Data Producer	Any person who creates the data in the application / system (e.g., data entry) and enables it to be ready for consumption. Example - Data owner, data scientist, data engineer
Data Profiling	The method used to gain more understanding of the data
Data Quality Index (DQI)	The index used to indicate how good the data quality is against the data quality dimension

GLOSSARY OF TERMS (2/2) (BY ALPHABETIC ORDER)

Term	Definition
Data Quality Process	Standardized operations for data quality governance across Group Data Zone and Monoline Data Zone
Data Quality Scorecard (DQ Scorecard)	The illustration that shows the quality of data
Data Quality Standard	Agreed definition, format and representation to unify approach to select and process data, and examine data quality
Data Set	A collection of data points related to a particular subject area
Data Subject Area	The logical grouping of available data sets such as customer, product, marketing etc.
Derived Data	New attributes derived from existing data to provide insights - this could be based on specific business rules or using analytics algorithms
Personally Identifiable Information (PII)	This data can identify data subjects which are direct data or indirect data. It is currently included in Business Metadata and Technical Metadata to provide PDPA information: PII, Non-PII.
Preventive Action	A long-term action to systematically prevent (re-)occurrence of certain data governance issue(s)
Remediation Plan	A plan to solve Data Quality issue. There are 2 types of remediation plan: Corrective and Preventive.
Service Agreement	Written contract between DataX and Monolines that outlines services to be performed by DataX, responsibilities of both parties, and payment(s) to be made by Monolines

Appendix B: Data Quality Management Detailed Processes

WE HAVE DEFINED DETAILED ACTIVITIES FOR 6 KEY PROCESSES TO GUIDE DATA QUALITY MGMT. ACROSS DATAx & MONOLINES (1/2)

Phase	#	Process Name	Description	Roles Involved *	
				Group Data Zone	Monoline Data Zone
Define / Refine	1	Critical Data Element (CDE) Identification / Maintenance and Data Quality (DQ) Dimension Mapping Process	Process to identify / maintain (and prioritize, <i>optional</i>) CDE, and map to DQ dimension for preliminary assessment	• DataX DQ Team • DataX Business Data Steward • DataX Data Owner • DataX Technical Data Steward	• Monoline DQ Team • Monoline Business Data Steward • Monoline Data Owner • Monoline Technical Data Steward
Discover	2	DQ Business Rules Identification and Refinement Process	Process to run data profiling to understand the data, and develop the fundamental and full list of DQ business rules	• DataX DQ Team • DataX Business Data Steward • DataX Data Owner • DataX Technical Data Steward	• Monoline DQ Team • Monoline Business Data Steward • Monoline Data Owner • Monoline Technical Data Steward
Diagnose	3	DQ Scorecard Generation and Analysis Process	Process to generate and analyse DQ Scorecard, and communicate the findings with Data Owner	• DataX DQ Team • DataX Business Data Steward • DataX Data Owner • DataX Technical Data Steward	• Monoline DQ Team • Monoline Business Data Steward • Monoline Data Owner • Monoline Technical Data Steward

DATA QUALITY MANAGEMENT DETAILED PROCESSES: OVERVIEW

WE HAVE DEFINED DETAILED ACTIVITIES FOR 6 KEY PROCESSES TO GUIDE DATA QUALITY MGMT. ACROSS DATAx & MONOLINES (2/2)

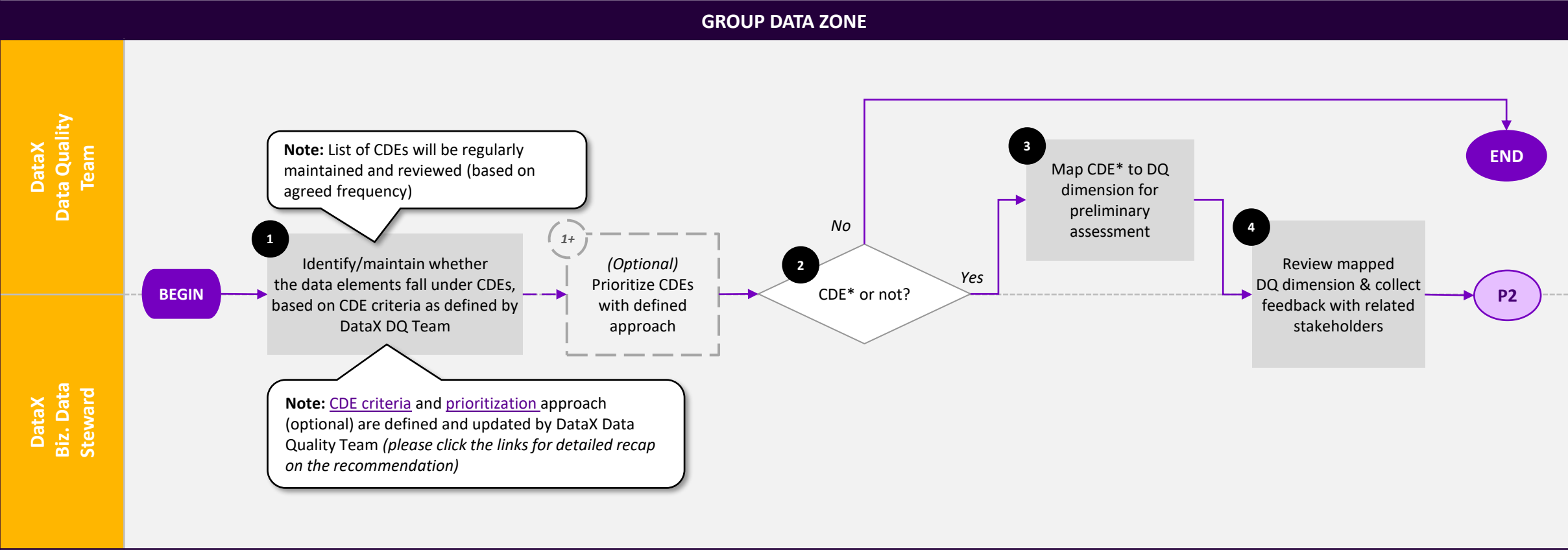
Phase	#	Process Name	Description	Roles Involved *	
				Group Data Zone	Monoline Data Zone
Remediate	4	DQ Issue Analysis Process	Process to conduct root-cause and impact analyses on the prioritized DQ issues, and communicate the results with data owner	• DataX DQ Team • DataX Business Data Steward • DataX Data Owner • DataX Technical Data Steward	• Monoline DQ Team • Monoline Business Data Steward • Monoline Data Owner • Monoline Technical Data Steward
Monitor	5	DQ Issue Remediation Process	Process to develop, confirm and execute remediation plan of DQ issues(s), including corrective and preventive actions	• DataX DQ Team • DataX Business Data Steward • DataX Data Owner • DataX Technical Data Steward • Identified stakeholders	• Monoline DQ Team • Monoline Business Data Steward • Monoline Data Owner • Monoline Technical Data Steward • Identified stakeholders
Monitor	6	DQ Scorecard Monitoring	Process to monitor DQ scorecard, analyse and communicate on the findings from monitoring, and decide whether any DQ issues need to be fixed	• DataX DQ Team • DataX Business Data Steward • DataX Data Owner • DataX Technical Data Steward	• Monoline DQ Team • Monoline Business Data Steward • Monoline Data Owner • Monoline Technical Data Steward

* Including core and related roles; respective responsibilities are detailed in RACI map for each process

PROCESS 1 (GDZ) TO IDENTIFY/MAINTAIN CRITICAL DATA ELEMENT AND MAP TO DATA QUALITY DIMENSION

Prerequisite(s)

- Data are already provisioned into Group Data Zone



PROCESS 1 (GDZ) TO IDENTIFY/MAINTAIN CDE AND MAP TO DQ DIMENSION: DETAILED ACTIVITIES

GROUP DATA ZONE			
Step	Activity Name	Activity Owner	Detailed Description
1	Identify/maintain whether the data elements fall under CDEs, based on CDE criteria as defined by DataX Data Quality Team	DataX Data Quality (DQ) Team; DataX Business Data Steward	DataX DQ Team and Business Data Steward to decide whether the data elements are Critical Data Elements (CDEs) or not, as per the CDE criteria defined by DataX
1+	(Optional) Prioritize CDEs with defined approach	DataX DQ Team; DataX Business Data Steward	DataX DQ Team and Business Data Steward to decide, among identified CDEs, which to be prioritized, as per the approach defined by DataX
2	CDE* or not?	DataX DQ Team; DataX Business Data Steward	DataX DQ Team and Business Data Steward to decide whether the data element is (prioritized) CDE or not: • If yes, proceed to step 3 • If not, process ends
3	Map CDE* to DQ dimension for preliminary assessment	DataX DQ Team	DataX DQ Team to map (prioritized) CDE to DQ dimension by relevancy for preliminary assessment on data quality
4	Review mapped DQ dimension & collect feedback with related stakeholders	DataX DQ Team; DataX Business Data Steward	DataX DQ Team to review mapped DQ dimension with DataX Business Data Steward and collect feedback with related stakeholders (e.g., monoline biz data steward)

* If the optional step of prioritization is adopted, the CDE in Step 2 and 3 will refer to the prioritized CDE

PROCESS 1 (GDZ) TO IDENTIFY/MAINTAIN CDE AND MAP TO DQ DIMENSION: RACI MAP

RACI mapped to key activities of the process: R: Responsible; A: Accountable; C: Consulted; I: Informed

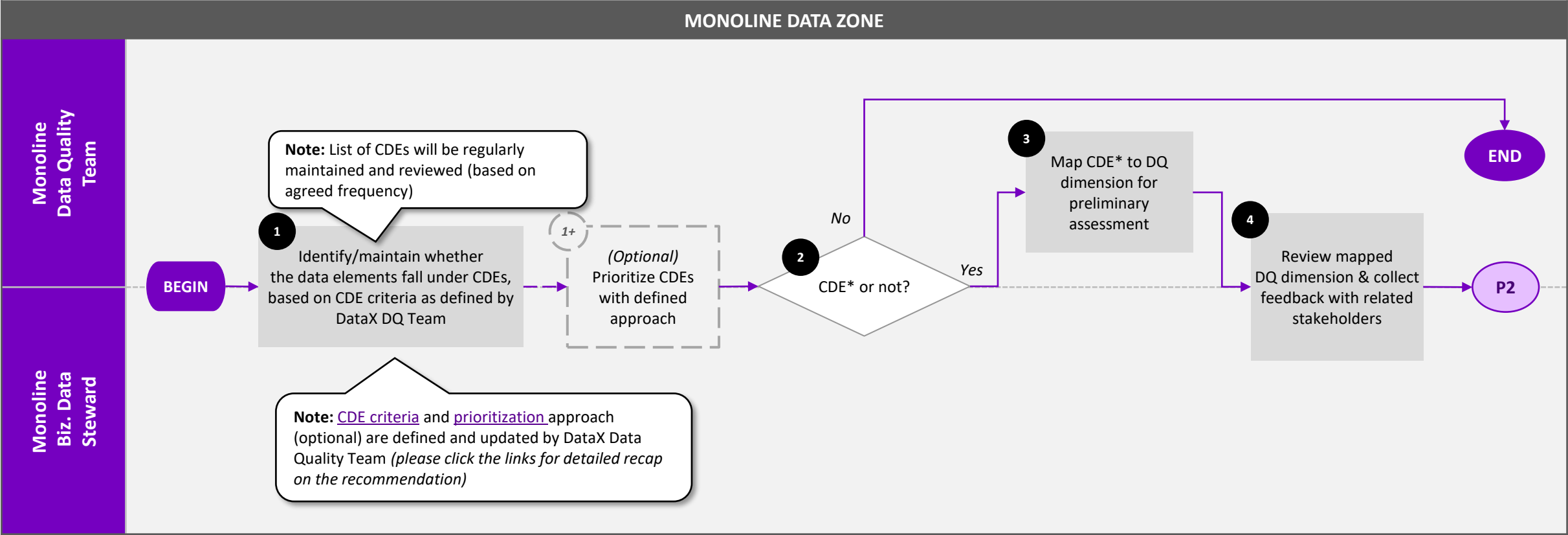
Step	Key Activity	GROUP DATA ZONE				
		DataX Data Quality (DQ) Team		DataX Biz. Data Steward	DataX Data Owner	DataX Technical Data Steward
		DQ Officer	Head of DQ			
1	Identify/maintain whether the data elements fall under CDEs, based on CDE criteria as defined by DataX Data Quality Team	R	A	R	A	I
1+	(Optional) Prioritize CDEs with defined approach	R	A	R	A	I
2	CDE* or not?	R	A	R	A	I
3	Map CDE* to DQ dimension for preliminary assessment	R	A	C	C	I
4	Review mapped DQ dimension & collect feedback with related stakeholders	R	A	R	A	C

* If the optional step of prioritization is adopted, the CDE in Step 2 and 3 will refer to the prioritized CDE

PROCESS 1 (MDZ) TO IDENTIFY/MAINTAIN CRITICAL DATA ELEMENT AND MAP TO DATA QUALITY DIMENSION

Prerequisite(s)

- Data are already provisioned into Monoline Data Zone



PROCESS 1 (MDZ) TO IDENTIFY/MAINTAIN CDE AND MAP TO DQ DIMENSION: DETAILED ACTIVITIES

MONOLINE DATA ZONE			
Step	Activity Name	Activity Owner	Detailed Description
1	Identify/maintain whether the data elements fall under CDEs, based on CDE criteria as defined by DataX Data Quality Team	Monoline Data Quality (DQ) Team; Monoline Business Data Steward	Monoline DQ Team and Business Data Steward to decide whether the data elements are Critical Data Elements (CDEs) or not, as per the CDE criteria defined by DataX
1+	(Optional) Prioritize CDEs with defined approach	Monoline DQ Team; Monoline Business Data Steward	Monoline DQ Team and Business Data Steward to decide, among identified CDEs, which to be prioritized, as per the approach defined by DataX
2	CDE* or not?	Monoline DQ Team; Monoline Business Data Steward	Monoline DQ Team and Business Data Steward to decide whether the data element is (prioritized) CDE or not: • If yes, proceed to step 3 • If not, process ends
3	Map CDE* to DQ dimension for preliminary assessment	Monoline DQ Team	Monoline Data Quality Team to map (prioritized) CDE to DQ dimension by relevancy for preliminary assessment on data quality
4	Review mapped DQ dimension & collect feedback with related stakeholders	Monoline DQ Team; Monoline Business Data Steward	Monoline DQ Team to review mapped DQ dimension with Monoline Business Data Steward and collect feedback with related stakeholders

* If the optional step of prioritization is adopted, the CDE in Step 2 and 3 will refer to the prioritized CDE

PROCESS 1 (MDZ) PROCESS TO IDENTIFY/MAINTAIN CDE AND MAP TO DQ DIMENSION: RACI MAP

RACI mapped to key activities of the process: R: Responsible; A: Accountable; C: Consulted; I: Informed

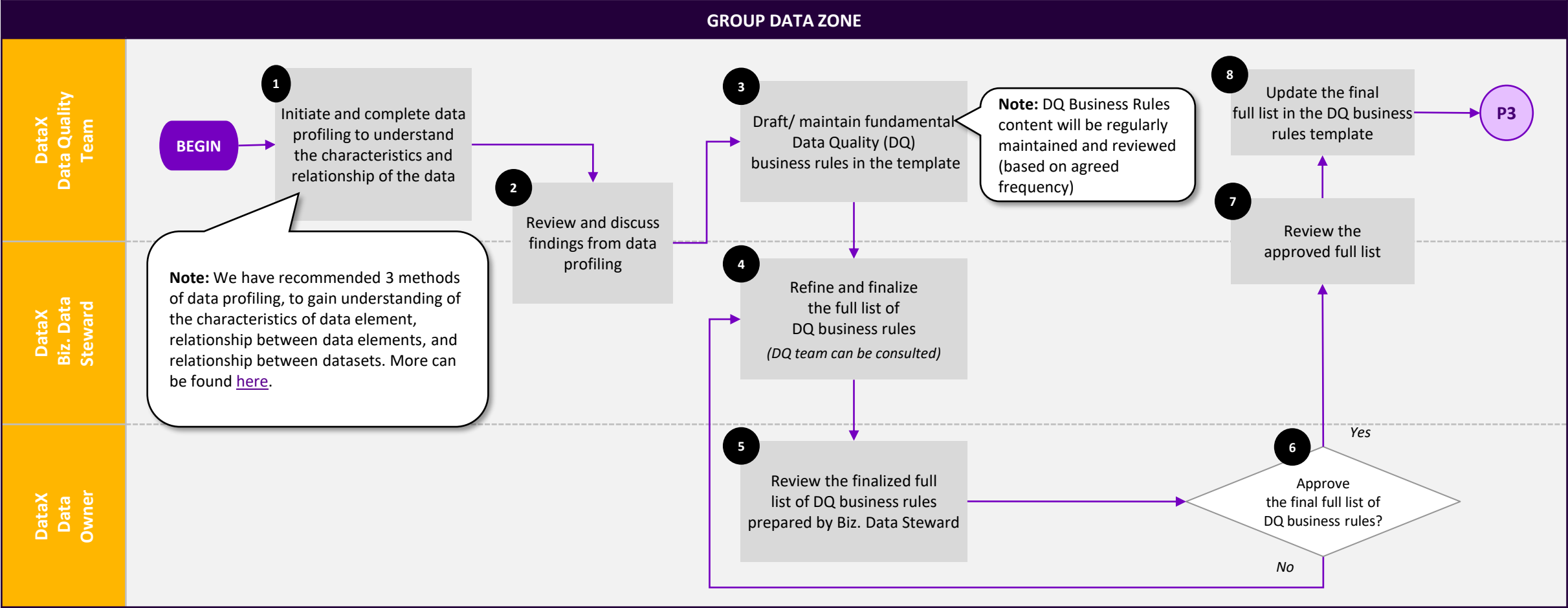
Step	Key Activity	MONOLINE DATA ZONE				
		Monoline Data Quality (DQ) Team		Monoline Biz. Data Steward	Monoline Data Owner	Monoline Technical Data Steward
		DQ Officer	Head of DQ			
1	Identify/maintain whether the data elements fall under CDEs, based on CDE criteria as defined by DataX Data Quality Team	R	A	R	A	I
1+	(Optional) Prioritize CDEs with defined approach	R	A	R	A	I
2	CDE* or not?	R	A	R	A	I
3	Map CDE* to DQ dimension for preliminary assessment	R	A	C	C	I
4	Review mapped DQ dimension & collect feedback with related stakeholders	R	A	R	A	C

* If the optional step of prioritization is adopted, the CDE in Step 2 and 3 will refer to the prioritized CDE

PROCESS 2 (GDZ) TO IDENTIFY AND REFINE DATA QUALITY BUSINESS RULE

Prerequisite(s)

- CDEs are already defined



PROCESS 2 (GDZ) TO IDENTIFY AND REFINE DATA QUALITY BUSINESS RULE: DETAILED ACTIVITIES

GROUP DATA ZONE			
Step	Activity Name	Activity Owner	Detailed Description
1	Initiate and complete data profiling to understand the characteristics and relationship of the data	DataX DQ Team	DataX Data Quality Team to run data profiling to gain understanding of the characteristics and relationship of the data
2	Review and discuss findings from data profiling	DataX DQ Team; DataX Business Data Steward	DataX Data Quality Team and Business Data Steward to review and discuss the findings from data profiling
3	Draft/ maintain fundamental Data Quality (DQ) business rules in the template	DataX DQ Team	DataX Data Quality Team to develop and provide fundamental DQ business rules
4	Refine and finalize the full list of DQ business rules	DataX Business Data Steward	DataX Business Data Steward to refine the fundamental DQ business rules prepared by DQ team and develop/update the final full list of DQ business rules
5	Review the finalized full list of DQ business rules prepared by Biz. Data Steward	DataX Data Owner	DataX Data Owner to review the finalized full list of DQ business rules developed/updated by the Business Data Steward
6	Approve the final full list of DQ business rules?	DataX Data Owner	DataX Data Owner to decide whether to approve the finalized full list of DQ business rules: • If yes, proceed to step 7 • If not, proceed to step 4 (i.e., DataX Biz. Data Steward to update the full list)
7	Review the approved full list	DataX DQ Team; DataX Business Data Steward	DataX Data Quality Team and Business Data Steward to review the final full list of DQ business rules
8	Update the final full list in the DQ business rules template	DataX DQ Team	DataX DQ Team to update the finalized full list approved by the Data Owner in the DQ business rules template

PROCESS 2 (GDZ) TO IDENTIFY AND REFINE DATA QUALITY BUSINESS RULE: RACI MAP

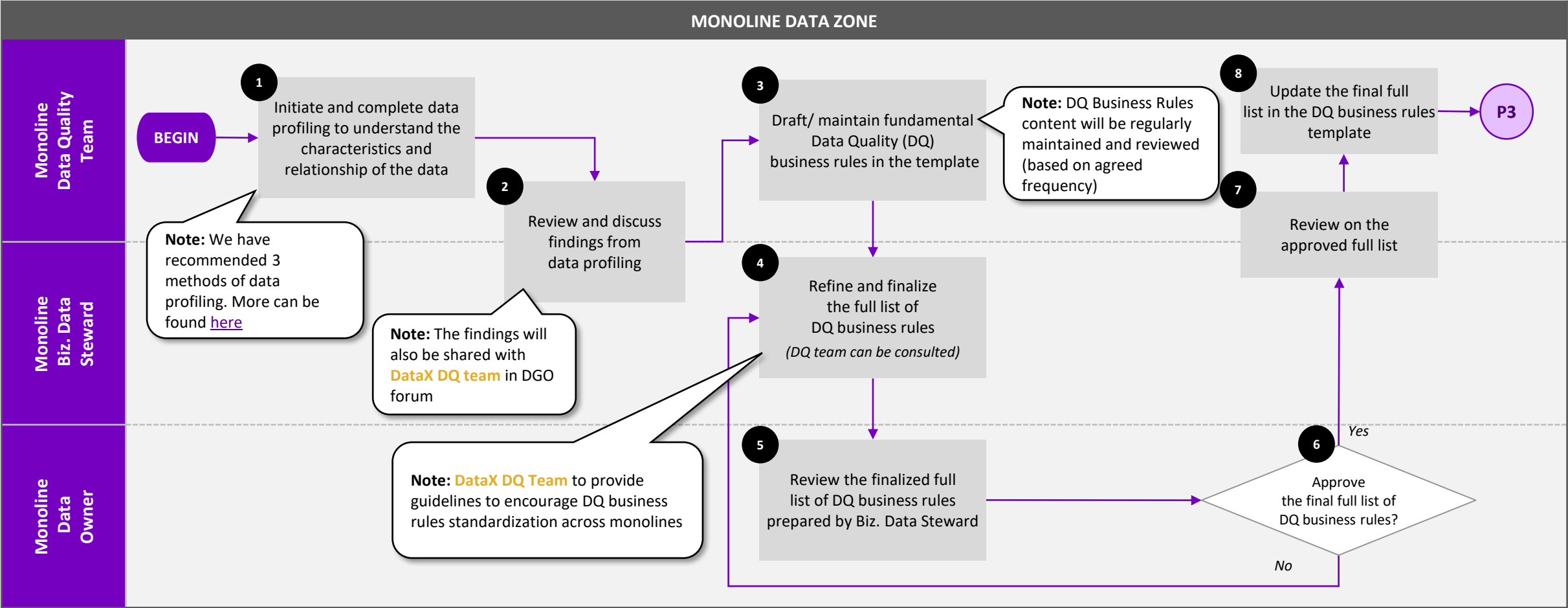
RACI mapped to key activities of the process: R: Responsible; A: Accountable; C: Consulted; I: Informed

Step	Key activity	GROUP DATA ZONE				
		DataX Data Quality (DQ) Team		DataX Biz. Data Steward	DataX Data Owner	DataX Technical Data Steward
		DQ Officer	Head of DQ			
1	Initiate and complete data profiling to understand the characteristics and relationship of the data	R	A	C	C	C
2	Review and discuss findings from data profiling	R	A	R	A	I
3	Draft/ maintain fundamental Data Quality (DQ) business rules in the template	R	A	I	I	I
4	Refine and finalize the full list of DQ business rules	C	I	R	I	C
5	Review the finalized full list of DQ business rules prepared by Biz. Data Steward	C	I	C	R, A	I
6	Approve the final full list of DQ business rules?	I	I	I	R, A	I
7	Review the approved full list	R	A	R	A	I
8	Update the final full list in the DQ business rules template	R	A	I	I	C

PROCESS 2 (MDZ) TO IDENTIFY AND REFINE DATA QUALITY BUSINESS RULE

Prerequisite(s)

- CDEs are already defined



PROCESS 2 (MDZ) TO IDENTIFY AND REFINE DATA QUALITY BUSINESS RULE: DETAILED ACTIVITIES

MONOLINE DATA ZONE			
Step	Activity Name	Activity Owner	Detailed Description
1	Initiate and complete data profiling to understand the characteristics and relationship of the data	Monoline DQ Team	Monoline Data Quality Team to run data profiling to gain understanding of the characteristics and relationship of the data
2	Review and discuss findings from data profiling	Monoline DQ Team; Monoline Business Data Steward	Monoline Data Quality Team and Business Data Steward to review and discuss the findings from data profiling
3	Draft/ maintain fundamental Data Quality (DQ) business rules in the template	Monoline DQ Team	Monoline Data Quality Team to develop and provide fundamental DQ business rules
4	Refine and finalize the full list of DQ business rules	Monoline Business Data Steward	Monoline Business Data Steward to refine the fundamental DQ business rules prepared by DQ team and develop/update the final full list of DQ business rules
5	Review the finalized full list of DQ business rules prepared by Biz. Data Steward	Monoline Data Owner	Monoline Data Owner to review the finalized full list of DQ business rules developed/updated by the Business Data Steward
6	Approve the final full list of DQ business rules?	Monoline Data Owner	Monoline Data Owner to decide whether to approve the finalized full list of DQ business rules: • If yes, proceed to step 7 • If not, proceed to step 4 (i.e., Monoline Biz. Data Steward to update the full list)
7	Review the approved full list	Monoline DQ Team; Monoline Business Data Steward	Monoline Data Quality Team and Business Data Steward to review the final full list of DQ business rules
8	Update the final full list in the DQ business rules template	Monoline DQ Team	Monoline DQ Team to update the finalized full list approved by the Data Owner in the DQ business rules template

PROCESS 2 (MDZ) TO IDENTIFY AND REFINE DATA QUALITY BUSINESS RULE: RACI MAP

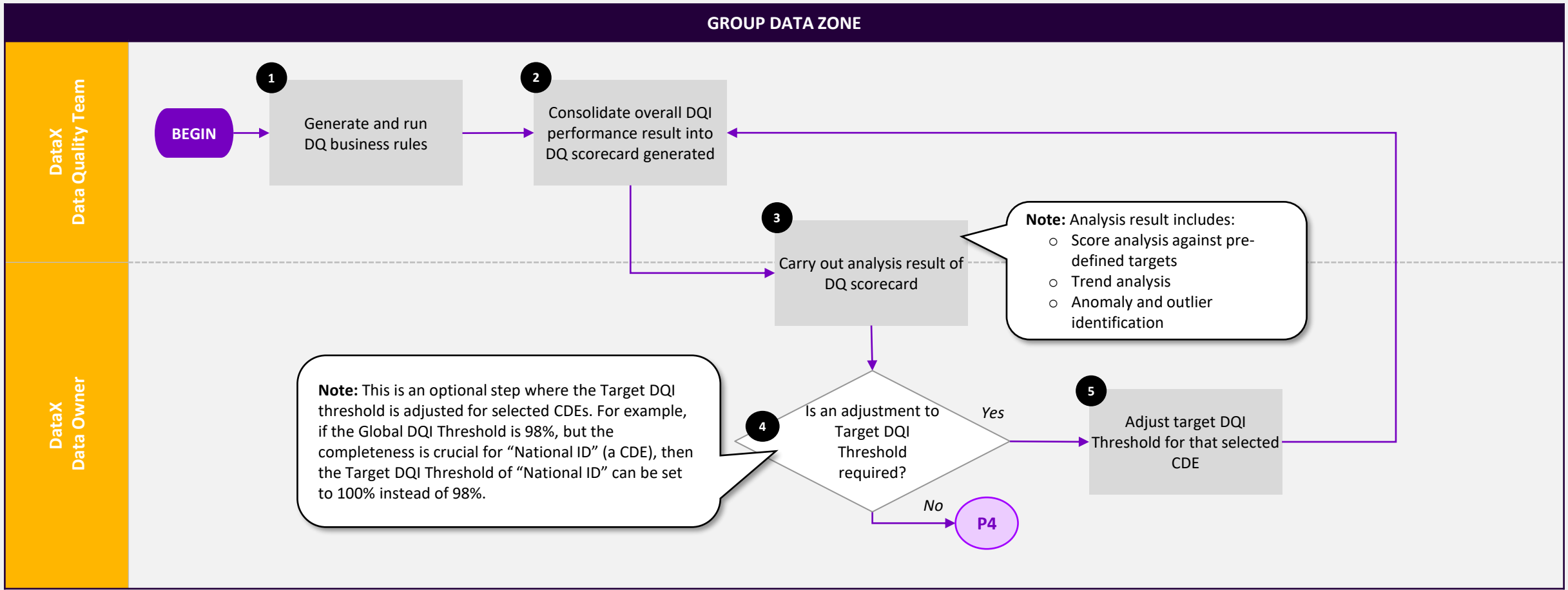
RACI mapped to key activities of the process: R: Responsible; A: Accountable; C: Consulted; I: Informed

Step	Key activity	MONOLINE DATA ZONE				
		Monoline Data Quality (DQ) Team		Monoline Biz. Data Steward	Monoline Data Owner	Monoline Technical Data Steward
		DQ Officer	Head of DQ			
1	Initiate and complete data profiling to understand the characteristics and relationship of the data	R	A	C	C	C
2	Review and discuss findings from data profiling	R	A	R	A	I
3	Draft/ maintain fundamental Data Quality (DQ) business rules in the template	R	A	I	I	I
4	Refine and finalize the full list of DQ business rules	C	I	R	I	C
5	Review the finalized full list of DQ business rules prepared by Biz. Data Steward	C	I	C	R, A	I
6	Approve the final full list of DQ business rules?	I	I	I	R, A	I
7	Review the approved full list	R	A	R	A	I
8	Update the final full list in the DQ business rules template	R	A	I	I	C

PROCESS 3 (GDZ) TO GENERATE AND ANALYZE DQ SCORECARD

Prerequisite(s)

- Business rules for data in GDZ have been defined/ refined and implemented in the tool
- DQ Scorecard in GDZ has been built and updated according to the latest business rules



PROCESS 3 (GDZ) TO GENERATE AND ANALYZE DQ SCORECARD: DETAILED ACTIVITIES

GROUP DATA ZONE			
Step	Activity Name	Activity Owner	Detailed Description
1	Generate and run DQ business rules	DataX Data Quality (DQ) Team	DataX DQ Team to apply and run DQ business rule and generate the DQ scorecard via the DQ tool once the DQ business rules are implemented and the DQ Scorecard is built. Additionally, the timeliness dimension will be used to measure timeliness at Data Set level.
2	Consolidate overall DQI performance result into DQ scorecard generated	DataX DQ Team	After DQ scorecard is generated, DataX DQ Team to consolidate the overall result from DQ scorecard
3	Carry out analysis result of DQ scorecard	DataX DQ Team, DataX Data Owner	DataX DQ team to analyze the result of the DQ scorecard generated with the following methods (non-exhaustive): • Score analysis against pre-defined targets • Trend analysis • Anomaly and outlier identification
4	Is an adjustment to Target DQI Threshold required?	DataX Data Owner	DataX Data Owner to decide if an adjustment to the Target DQI threshold is required by considering the CDE classification, context and use case of the data • If yes, proceed to Step 5 • If no, proceed to Process 4 (to identify whether there is any DQ issue)
5	Adjust target DQI Threshold for that selected CDE	DataX Data Owner	If a target DQI threshold is required, DataX Data Owner to determine the DQI threshold to be applied and adjust the Target DQI threshold accordingly

PROCESS 3 (GDZ) TO GENERATE AND ANALYZE DQ SCORECARD: RACI MAP

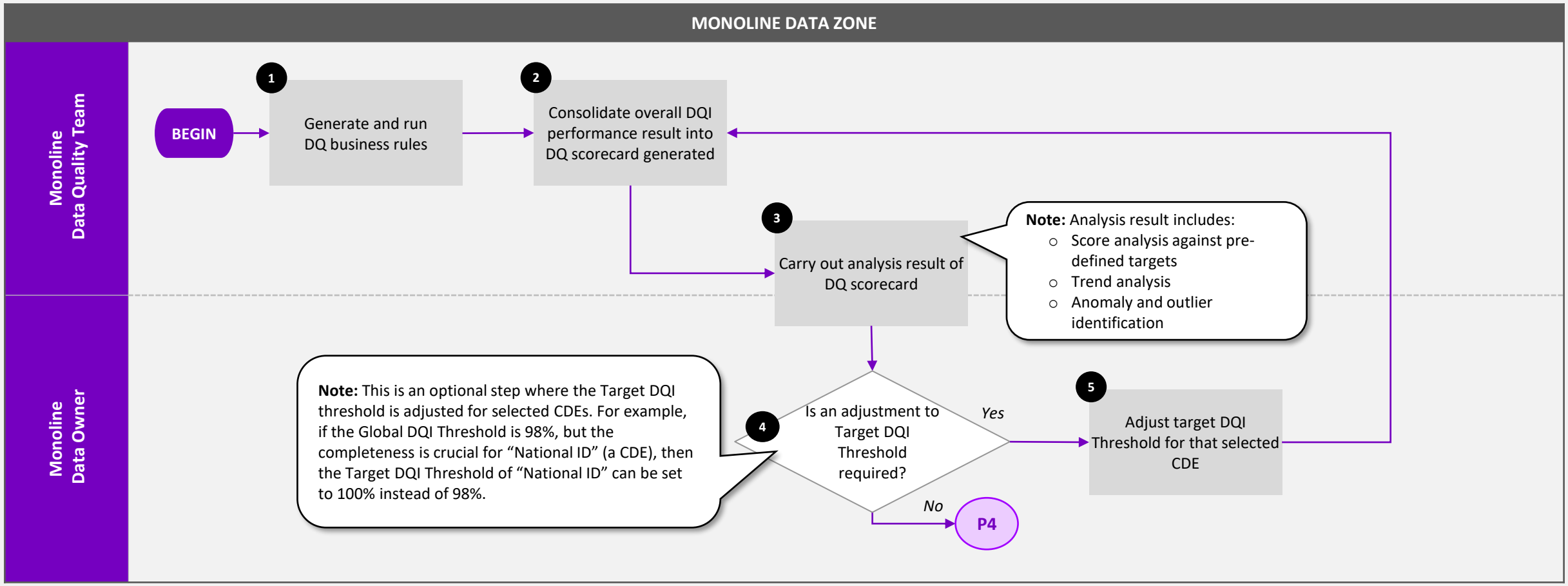
RACI mapped to key activities of the process: R: Responsible; A: Accountable; C: Consulted; I: Informed

Step	Key Activity	GROUP DATA ZONE				
		DataX Data Quality (DQ) Team		DataX Biz. Data Steward	DataX Data Owner	DataX Technical Data Steward
		DQ Officer	Head of DQ			
1	Generate and run DQ business rules	R	A	I	C	I
2	Consolidate overall DQI performance result into DQ scorecard generated	R	A	I	C	I
3	Carry out analysis result of DQ scorecard	R	A	I	R	I
4	Is an adjustment to Target DQI Threshold required?	C	A	C, I	R	C, I
5	Adjust target DQI Threshold for that selected CDE	C	A	C, I	R	C, I

PROCESS 3 (MDZ) TO GENERATE AND ANALYZE DQ SCORECARD

Prerequisite(s)

- Business rules for data in MDZ have been defined/ refined and implemented in the tool
- DQ Scorecard in MDZ has been built and updated according to the latest business rules



PROCESS 3 (MDZ) TO GENERATE AND ANALYZE DQ SCORECARD: DETAILED ACTIVITIES

GROUP DATA ZONE			
Step	Activity Name	Activity Owner	Detailed Description
1	Generate and run DQ business rules	Monoline Data Quality (DQ) Team	Monoline DQ Team to apply and run DQ business rule and generate the DQ scorecard via the DQ tool once the DQ business rules are implemented and the DQ Scorecard is built. Additionally, the timeliness dimension will be used to measure timeliness at Data Set level.
2	Consolidate overall DQI performance result into DQ scorecard generated	Monoline DQ Team	After DQ scorecard is generated, Monoline DQ Team to consolidate the overall result from DQ scorecard
3	Carry out analysis result of DQ scorecard	Monoline DQ Team, Monoline Data Owner	Monoline DQ team to analyze the result of the DQ scorecard generated with the following methods (non-exhaustive): - Score analysis against pre-defined targets - Trend analysis - Anomaly and outlier identification
4	Is an adjustment to Target DQI Threshold required?	Monoline Data Owner	Monoline Data Owner to decide if an adjustment to the Target DQI threshold is required by considering the CDE classification, context and use case of the data - If yes, proceed to Step 5 - If no, proceed to Process 4 (to identify whether there is any DQ issue)
5	Adjust target DQI Threshold for that selected CDE	Monoline Data Owner	If a target DQI threshold is required, Monoline Data Owner to determine the DQI threshold to be applied and adjust the Target DQI threshold accordingly

PROCESS 3 (MDZ) TO GENERATE AND ANALYZE DQ SCORECARD: RACI MAP

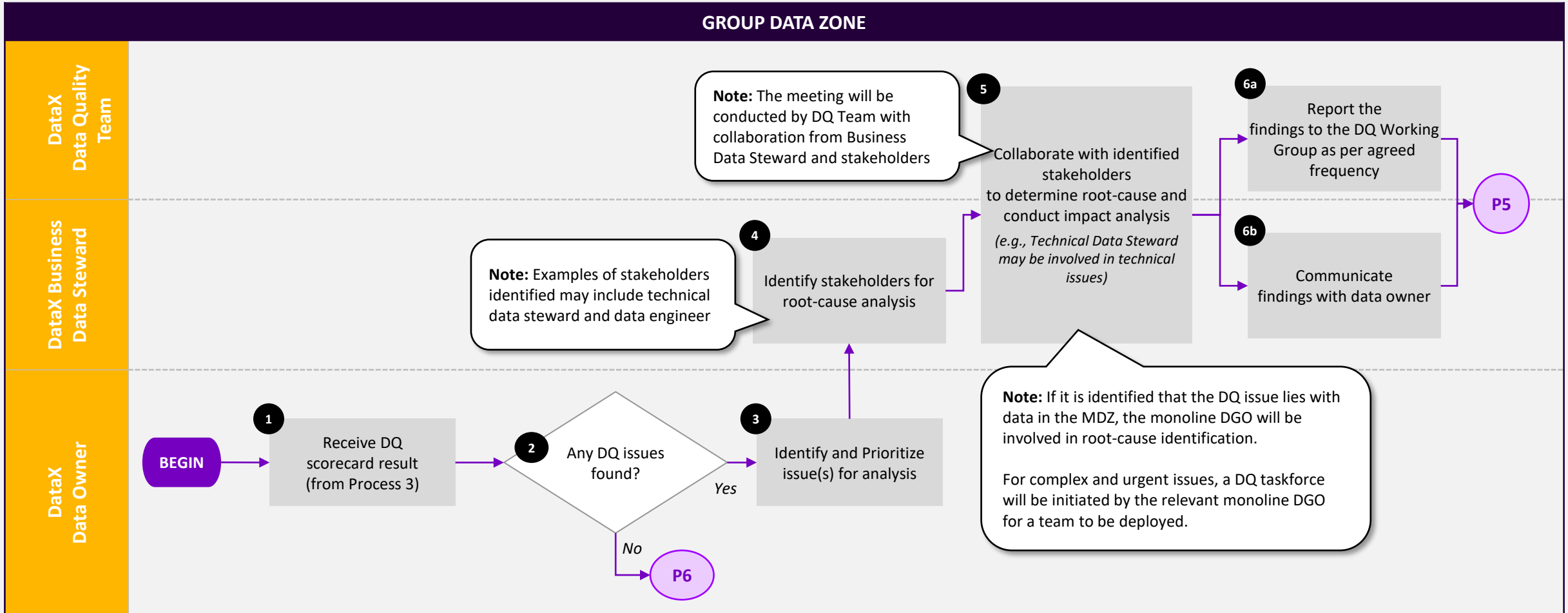
RACI mapped to key activities of the process: R: Responsible; A: Accountable; C: Consulted; I: Informed

Step	Key Activity	MONOLINE DATA ZONE				
		Monoline Data Quality (DQ) Team		Monoline Biz. Data Steward	Monoline Data Owner	Monoline Technical Data Steward
		DQ Officer	Head of DQ			
1	Generate and run DQ business rules	R	A	I	C	I
2	Consolidate overall DQI performance result into DQ scorecard generated	R	A	I	C	I
3	Carry out analysis result of DQ scorecard	R	A	I	R	I
4	Is an adjustment to Target DQI Threshold required?	C	A	C, I	R	C, I
5	Adjust target DQI Threshold for that selected CDE	C	A	C, I	R	C, I

PROCESS 4 (GDZ) TO PRIORITIZE AND ANALYZE DQ ISSUES IDENTIFIED

Prerequisite(s)

- Scorecard analysis is also done (in Process 3)
- Issue is identified and added to data owner’s backlog



PROCESS 4 (GDZ) TO PRIORITIZE AND ANALYZE DQ ISSUES IDENTIFIED: DETAILED ACTIVITIES

GROUP DATA ZONE			
Step	Activity Name	Activity Owner	Detailed Description
1	Receive DQ scorecard result (from Process 3)	DataX Data Owner	DataX Data Owner to receive DQ scorecard result from Process 3
2	Any DQ issues found?	DataX Data Owner	If no DQ issues found, DataX Data Owner to proceed to Process 6 If DQ issues found from DQ scorecard, DataX Data Owner to proceed to step 3
3	Identify and Prioritize issue(s) for analysis	DataX Data Owner	DataX Data Owner to determine prioritized issue(s) for analysis based on the list of issues in the backlog and CDE classification of the impacted data
4	Identify stakeholders for root-cause analysis	DataX Business Data Steward	DataX Business Steward to identify stakeholders required to investigate the issue. Stakeholders may include data engineer, technical data steward etc.
5	Collaborate with identified stakeholders to determine root-cause and conduct impact analysis <i>(e.g., Technical Data Steward may be involved in technical issues)</i>	DataX Business Steward; DataX DQ Team	DataX DQ Team to conduct a joint meeting for DataX Business Steward and DataX Data Quality Team to jointly drive discussions to drill-down on DQ issues to identify how and where the issue stemmed from, and analyze its impact
6a	Report the findings to the DQ Working Group as per agreed frequency	DataX DQ Team	DataX Data Quality Team to report the findings from root-cause analysis and impact analysis to the DQ Working Group as per agreed frequency
6b	Communicate findings with data owner	DataX BusinessData Steward	DataX Data Steward to communicate findings from root-cause analysis and impact analysis with Data Owner

PROCESS 4 (GDZ) TO PRIORITIZE AND ANALYZE DQ ISSUES IDENTIFIED: RACI MAP

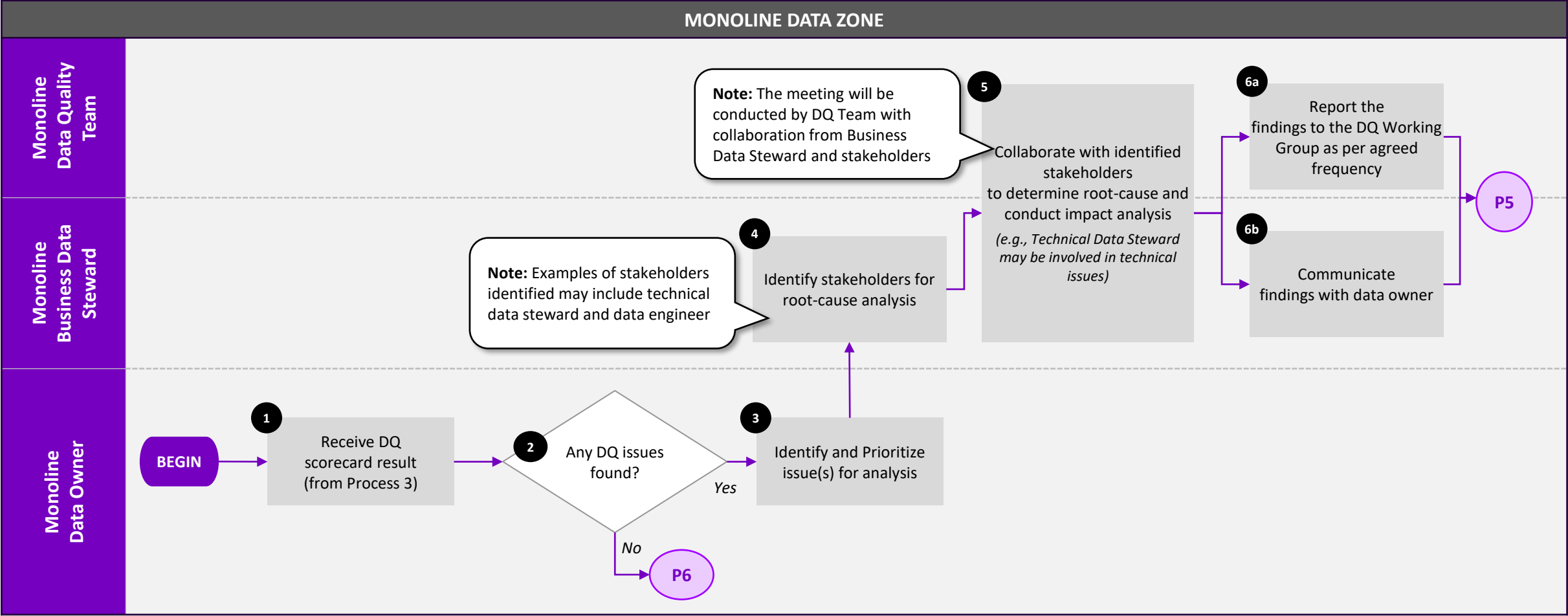
RACI mapped to key activities of the process: R: Responsible; A: Accountable; C: Consulted; I: Informed

Step	Key Activity	GROUP DATA ZONE				
		DataX Data Quality (DQ) Team		DataX Biz. Data Steward	DataX Data Owner	DataX Technical Data Steward
		DQ Officer	Head of DQ			
1	Receive DQ scorecard result (from Process 3)	C	C	I	A, R	I
2	Any DQ issues found?	C, I	C, I	I	A, R	I
3	Identify and Prioritize issue(s) for analysis	C, I	C, I	I	A, R	I
4	Identify stakeholders for root-cause analysis	I	I	R	A, I	C
5	Collaborate with identified stakeholders to determine root-cause and conduct impact analysis (e.g., Technical Data Steward may be involved in technical issues)	R	I	R	A, I	C
6a	Report the findings to the DQ Working Group as per agreed frequency	R	A	I	I	I
6b	Communicate findings with data owner	I	I	R	A	I

PROCESS 4 (MDZ) TO PRIORITIZE AND ANALYZE DQ ISSUES IDENTIFIED

Prerequisite(s)

- Scorecard analysis is also done (in Process 3)
- Issue is identified and added to data owner’s backlog



PROCESS 4 (MDZ) TO PRIORITIZE AND ANALYZE DQ ISSUES IDENTIFIED: DETAILED ACTIVITIES

MONOLINE DATA ZONE			
Step	Activity Name	Activity Owner	Detailed Description
1	Receive DQ scorecard result (from Process 3)	Monoline Data Owner	Monoline Data Owner to receive DQ scorecard result from Process 3
2	Any DQ issues found?	Monoline Data Owner	If no DQ issues found, Monoline Data Owner to proceed to Process 6 If DQ issues found from DQ scorecard, Monoline Data Owner to proceed to step 3
3	Identify and Prioritize issue(s) for analysis	Monoline Data Owner	Monoline Data Owner to determine prioritized issue(s) for analysis based on the list of issues in the backlog and CDE classification of the impacted data
4	Identify stakeholders for root-cause analysis	Monoline Business Data Steward	Monoline Business Steward to identify stakeholders required to investigate the issue. Stakeholders may include data engineer, technical data steward etc.
5	Collaborate with identified stakeholders to determine root-cause and conduct impact analysis <i>(e.g., Technical Data Steward may be involved in technical issues)</i>	Monoline Business Steward; Monoline DQ Team	Monoline DQ Team to conduct a joint meeting for Monoline Business Steward and Monoline Data Quality Team to jointly drive discussions to drill-down on DQ issues to identify how and where the issue stemmed from, and analyze its impact
6a	Report the findings to the DQ Working Group as per agreed frequency	Monoline DQ Team	Monoline Data Quality Team to report the findings from root-cause analysis and impact analysis to the DQ Working Group as per agreed frequency
6b	Communicate findings with data owner	Monoline Business Data Steward	Monoline Data Steward to communicate findings from root-cause analysis and impact analysis with Data Owner

PROCESS 4 (MDZ) TO PRIORITIZE AND ANALYZE DQ ISSUES IDENTIFIED: RACI MAP

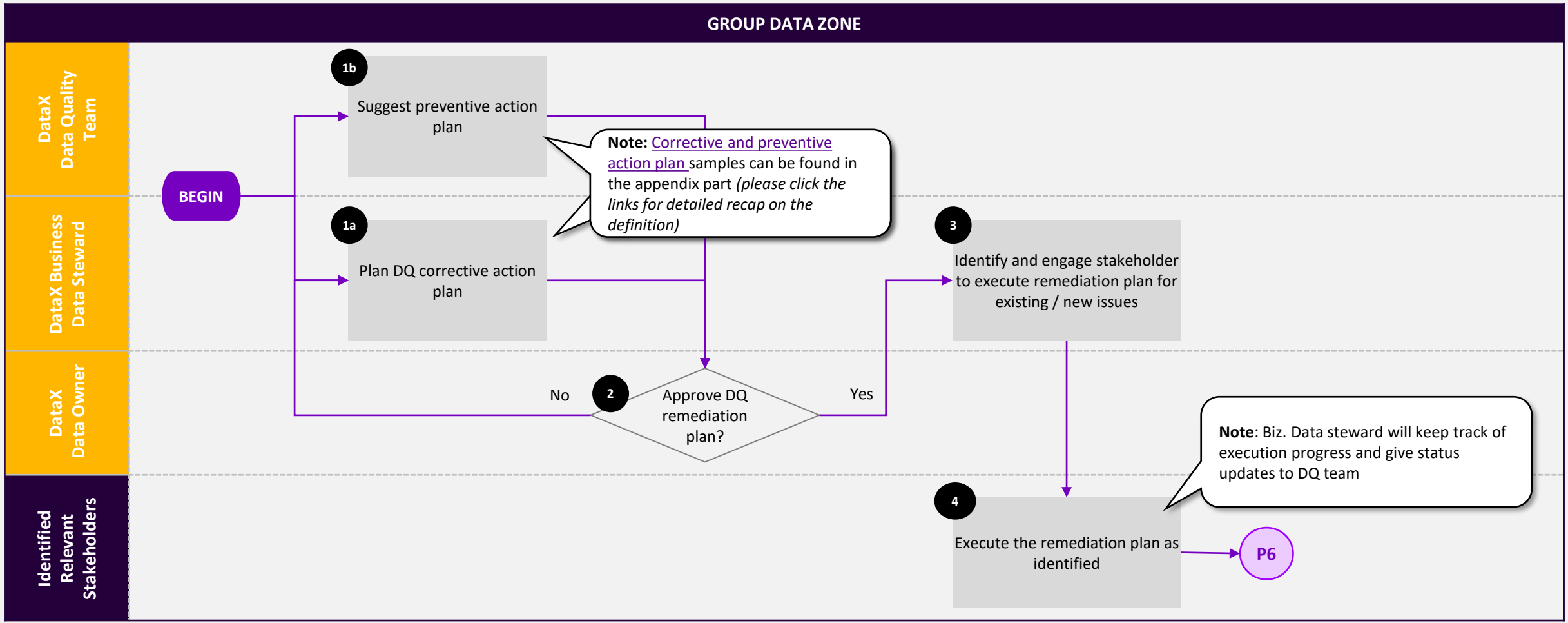
RACI mapped to key activities of the process: R: Responsible; A: Accountable; C: Consulted; I: Informed

Step	Key Activity	Monoline DATA ZONE				
		Monoline Data Quality (DQ) Team		Monoline Biz. Data Steward	Monoline Data Owner	Monoline Technical Data Steward
		DQ Officer	Head of DQ			
1	Receive DQ scorecard result (from Process 3)	C	C	I	A, R	I
2	Any DQ issues found?	C, I	C, I	I	A, R	I
3	Identify and Prioritize issue(s) for analysis	C, I	C, I	I	A, R	I
4	Identify stakeholders for root-cause analysis	I	I	R	A, I	C
5	Collaborate with identified stakeholders to determine root-cause and conduct impact analysis <i>(e.g., Technical Data Steward may be involved in technical issues)</i>	R	I	R	A, I	C
6a	Report the findings to the DQ Working Group as per agreed frequency	R	A	I	I	I
6b	Communicate findings with data owner	I	I	R	A	I

PROCESS 5 (GDZ) TO DEVELOP, EXECUTE, AND MONITOR REMEDIATION PLAN

Prerequisite(s)

- Data Quality issues in GDZ are analyzed



PROCESS 5 (GDZ) TO DEVELOP, EXECUTE, AND MONITOR REMEDIATION PLAN: DETAILED ACTIVITIES

GROUP DATA ZONE			
Step	Activity Name	Activity Owner	Detailed Description
1a	Plan DQ corrective action plan	DataX Business Data Steward	While DataX DQ Team is suggesting preventive action plan, DataX Business Data Steward to plan corrective action plan and provide the information in governance forums if it is required
1b	Suggest preventive action plan	DataX Data Quality (DQ) Team	DataX DQ Team to suggest preventive action plan to solve data quality issue
2	Approve DQ remediation plan?	DataX Data Owner	DataX Data Owner to approve the DQ remediation plan provided by DataX Business Data Steward and DataX DQ Team. If the corrective action plan is not approved, go back to step 1a. If preventive action plan is not approved, go back to step 1b If both corrective and preventive plan are approved, proceed to step 3.
3	Identify and engage stakeholder to execute remediation plan for existing / new issues	DataX Business Data Steward	DataX Business Data Steward to identify the relevant stakeholder for remediation plan (both preventive action and corrective action plan) and engage them to execute the plan
4	Execute the remediation plan as identified	Identified Relevant Stakeholder	Identified Relevant Stakeholder to execute the remediation to solve the data quality issue

PROCESS 5 (GDZ) TO MONITOR AND COMMUNICATE THE DATA QUALITY SCORECARD RESULTS : RACI MAP

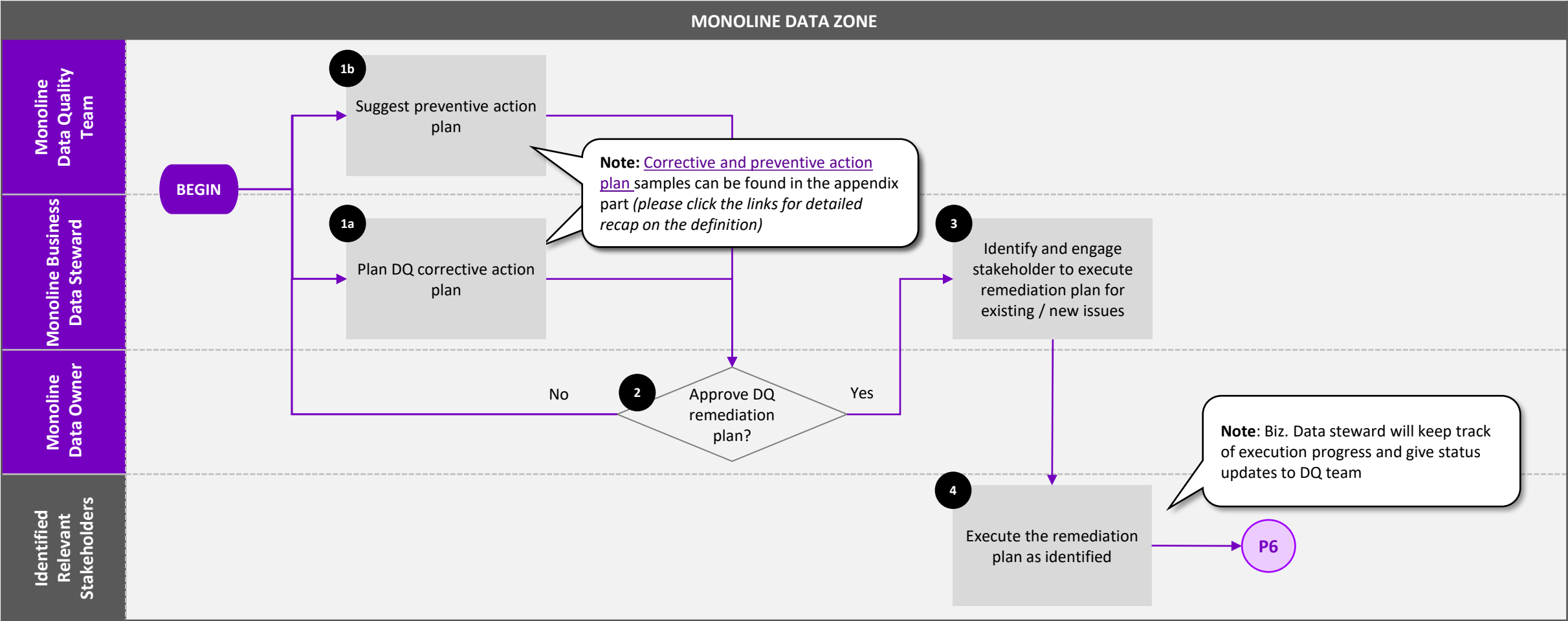
RACI mapped to key activities of the process: R: Responsible; A: Accountable; C: Consulted; I: Informed

Step	Key activity	GROUP DATA ZONE					
		DataX Data Quality (DQ) Team		DataX Biz. Data Steward	DataX Data Owner	DataX Technical Data Steward	Identified Stakeholder
		DQ Officer	Head of DQ				
1a	Plan DQ corrective action plan	I	I	R	A	C	-
1b	Suggest preventive action plan	R	A	I	I	C	-
2	Approve DQ remediation plan?	I	I	I	R, A	I	-
3	Identify and engage stakeholder to execute remediation plan for existing / new issues	C	C	R	A	C	I
4	Execute the remediation plan as identified	I	I	C	C	C	R, A

PROCESS 5 (MDZ) TO DEVELOP, EXECUTE, AND MONITOR REMEDIATION PLAN

Prerequisite(s)

- Data Quality issues in MDZ are analyzed



PROCESS 5 (MDZ) TO DEVELOP, EXECUTE, AND MONITOR REMEDIATION PLAN: DETAILED ACTIVITIES

MONOLINE DATA ZONE			
Step	Activity Name	Activity Owner	Detailed Description
1a	Plan DQ corrective action plan	Monoline Business Data Steward	While Monoline DQ Team is suggesting preventive action plan, Monoline Business Data Steward to plan corrective action plan and provide the information in governance forums if it is required
1b	Suggest preventive action plan	Monoline Data Quality (DQ) Team	Monoline DQ Team to suggest preventive action plan to solve data quality issue
2	Approve DQ remediation plan?	Monoline Data Owner	Monoline Data Owner to approve the DQ remediation plan provided by Monoline Business Data Steward and Monoline DQ Team. If the corrective action plan is not approved, go back to step 1a. If preventive action plan is not approved, go back to step 1b If both corrective and preventive plan are approved, proceed to step 3.
3	Identify and engage stakeholder to execute remediation plan for existing / new issues	Monoline Business Data Steward	Monoline Business Data Steward to identify the relevant stakeholder for remediation plan (both preventive action and corrective action plan) and engage them to execute the plan
4	Execute the remediation plan as identified	Identified Relevant Stakeholder	Identified Relevant Stakeholder to execute the remediation to solve the data quality issue

PROCESS 5 (MDZ) TO MONITOR AND COMMUNICATE THE DATA QUALITY SCORECARD RESULTS : RACI MAP

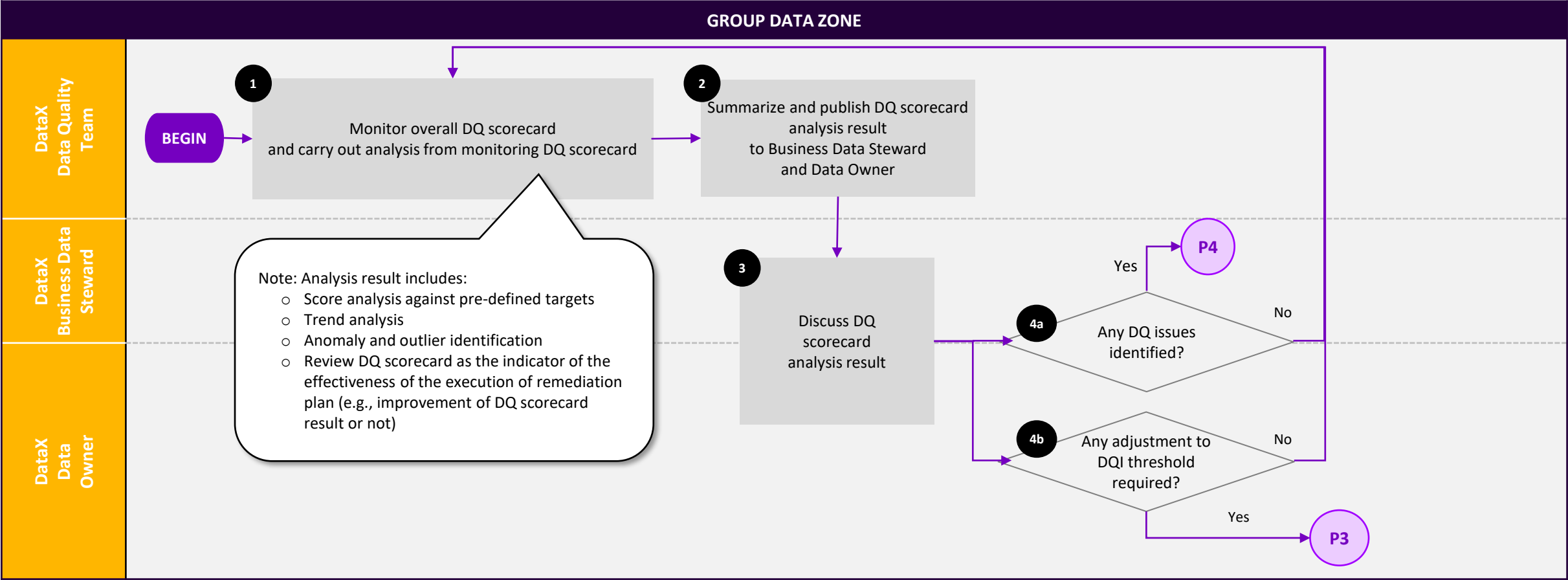
RACI mapped to key activities of the process: R: Responsible; A: Accountable; C: Consulted; I: Informed

Step	Key activity	MONOLINE DATA ZONE					
		DataX Data Quality (DQ) Team		DataX Biz. Data Steward	DataX Data Owner	DataX Technical Data Steward	Identified Stakeholder
		DQ Officer	Head of DQ				
1a	Plan DQ corrective action plan	I	I	R	A	C	-
1b	Suggest preventive action plan	R	A	I	I	C	-
2	Approve DQ corrective action plan?	I	I	I	R, A	I	-
3	Identify and engage stakeholder to execute remediation plan for existing / new issues	C	C	R	A	C	I
4	Execute the remediation plan as identified	I	I	C	C	C	R, A

PROCESS 6 (GDZ) TO MONITOR AND COMMUNICATE THE DATA QUALITY SCORECARD RESULTS

Prerequisite(s)

- Data Quality Scorecard is generated
- Data Quality Tool / Scorecard is established and updated



PROCESS 6 (GDZ) TO MONITOR AND COMMUNICATE THE DATA QUALITY SCORECARD RESULTS: DETAILED ACTIVITIES

GROUP DATA ZONE			
Step	Activity Name	Activity Owner	Detailed Description
1	Monitor overall DQ scorecard, and carry out analysis from monitoring DQ scorecard	DataX Data Quality (DQ) Team	DataX DQ Team to monitor DQ scorecard periodically as frequency defined in metrics, and to carry out analysis on the result of DQ scorecard from monitoring
2	Summarize and publish DQ scorecard analysis result to Business Data Steward and Data Owner	DataX DQ Team	DataX DQ Team to summarize and publish the DQ scorecard result to DataX Business Data Steward and Data Owner for their further discussion and analysis
3	Discuss DQ scorecard analysis result	DataX Biz. Data Steward; DataX Data Owner	DataX Business Data Steward and Data Owner to discuss the analysis result of DQ scorecard
4a	Any DQ issues identified?	DataX Biz. Data Steward; DataX Data Owner	DataX Business Data Steward and Data Owner to identify whether any DQ issues need to be solved • If DQ issue is identified, proceed to Process 4 • If no DQ issue is identified, the process will continue back to Step 1
4b	Any adjustment to DQI threshold required?	DataX Data Owner	DataX Data Owner to identify whether any systematic adjustment are needed: • If adjustment is needed for DQI threshold, proceed to Process 3 • If no adjustment is needed, the process will continue back to Step 1

PROCESS 6 (GDZ) TO MONITOR AND COMMUNICATE THE DATA QUALITY SCORECARD RESULTS : RACI MAP

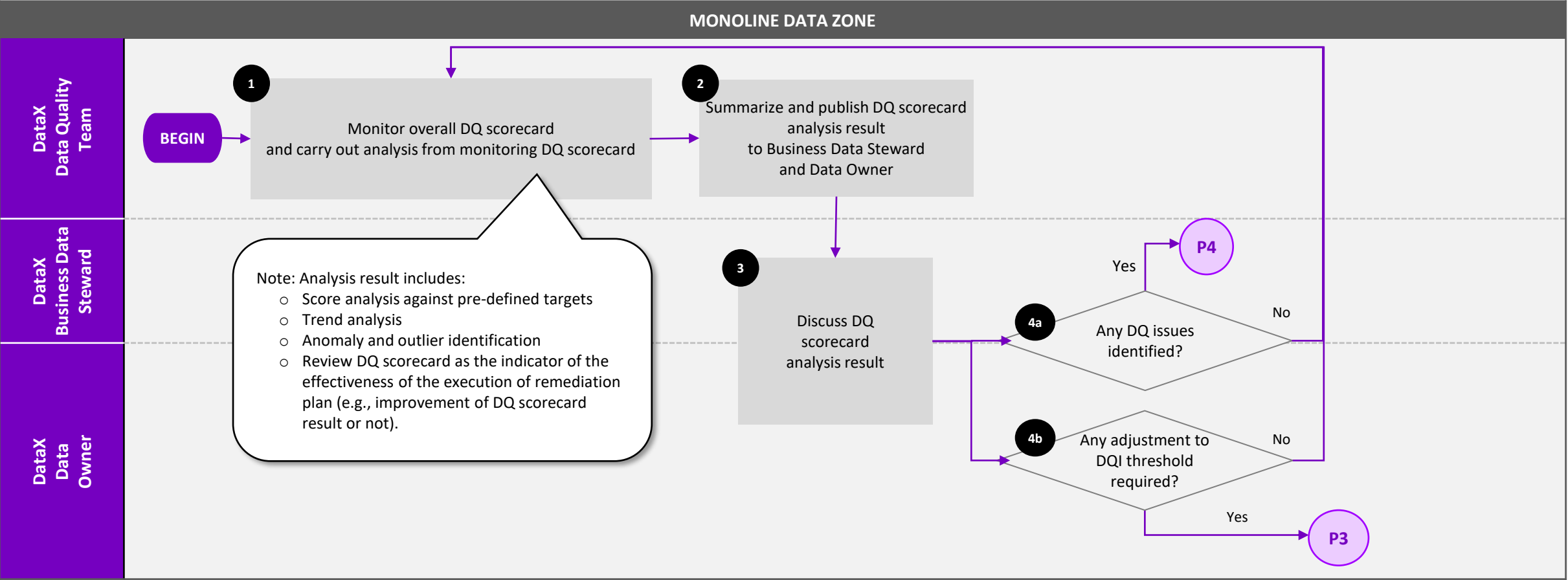
RACI mapped to key activities of the process: R: Responsible; A: Accountable; C: Consulted; I: Informed

Step	Key activity	GROUP DATA ZONE				
		DataX Data Quality (DQ) Team		DataX Biz. Data Steward	DataX Data Owner	DataX Technical Data Steward
		DQ Officer	Head of DQ			
1	Monitor overall DQ scorecard, and carry out analysis from monitoring DQ scorecard	R	A	I	I	I
2	Summarize and publish DQ scorecard analysis result to Business Data Steward and Data Owner	R	A	I	I	I
3	Discuss DQ scorecard analysis result	C	C	R	A	I
4a	Any DQ issues identified?	C	C	R	A	I
4b	Any adjustment to DQI threshold required?	C	C	I	R, A	I

PROCESS 6 (MDZ) TO MONITOR AND COMMUNICATE THE DATA QUALITY SCORECARD RESULTS

Prerequisite(s)

- Data Quality Scorecard is generated
- Data Quality Tool / Scorecard is established and updated



PROCESS 6 (MDZ) TO MONITOR AND COMMUNICATE THE DATA QUALITY SCORECARD RESULTS: DETAILED ACTIVITIES

MONOLINE DATA ZONE			
Step	Activity Name	Activity Owner	Detailed Description
1	Monitor overall DQ scorecard, and carry out analysis from monitoring DQ scorecard	Monoline Data Quality (DQ) Team	Monoline DQ Team to monitor DQ scorecard periodically as frequency defined in metrics and carry out analysis on the result of DQ scorecard from monitoring
2	Summarize and publish DQ scorecard analysis result to Business Data Steward and Data Owner	Monoline DQ Team	Monoline DQ Team to summarize and publish the DQ scorecard result to Monoline Business Data Steward and Data Owner for their further discussion and analysis
3	Discuss DQ scorecard analysis result	Monoline Biz. Data Steward; Monoline Data Owner	Monoline Business Data Steward and Data Owner to discuss the analysis result of DQ scorecard
4a	Any DQ issues identified?	Monoline Biz. Data Steward; Monoline Data Owner	Monoline Business Data Steward and Data Owner to identify whether any DQ issues need to be solved • If DQ issue is identified, proceed to Process 4 • If no DQ issue is identified, the process will continue back to Step 1
4b	Any adjustment to DQI threshold required?	DataX Data Owner	DataX Data Owner to identify whether any systematic adjustment are needed: • If adjustment is needed for DQI threshold, proceed to Process 3 • If no adjustment is needed, the process will continue back to Step 1

PROCESS 6 (MDZ) TO MONITOR AND COMMUNICATE THE DATA QUALITY SCORECARD RESULTS : RACI MAP

RACI mapped to key activities of the process: R: Responsible; A: Accountable; C: Consulted; I: Informed

Step	Key activity	MONOLINE DATA ZONE				
		Monoline Data Quality (DQ) Team		Monoline Biz. Data Steward	Monoline Data Owner	Monoline Technical Data Steward
		DQ Officer	Head of DQ			
1	Monitor overall DQ scorecard, and carry out analysis from monitoring DQ scorecard	R	A	I	I	I
2	Summarize and publish DQ scorecard analysis result to Business Data Steward and Data Owner	R	A	I	I	I
3	Discuss DQ scorecard analysis result	C	C	R	A	I
4a	Any DQ issues identified?	C	C	R	A	I
4b	Any adjustment to DQI threshold required?	C	C	I	R, A	I

Appendix C: Data Quality Management Key Metrics

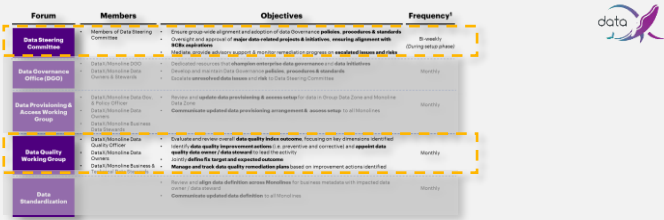
PROPOSED KEY METRICS TO DATAX DATA QUALITY TEAM

Related Phase	Related Process	#	Metric	Description	Example metrics*
Define/ Refine	Process 1: Critical Data Element (CDE) Identification / Maintenance and Data Quality (DQ) Dimension Mapping Process	1	List of CDEs Review and Maintenance	Frequency of reviewing list of CDEs to ensure they are up-to-date	Quarterly
Discover	Process 2: DQ Business Rules Identification and Refinement Process	2	DQ Business Rules Content Review and Maintenance	Frequency of reviewing DQ Business Rules content to ensure they are up-to-date	Quarterly
Remediate	Process 4: DQ Issue Analysis Process	3	% of DQ Issues Analyzed	% of issues with root-cause determined and impact analyzed (against total # of issues identified)	>= 80%
Monitor	Process 6: DQ Scorecard Monitoring	4	DQ Scorecard Monitoring Frequency	Frequency for DQ Team to regularly monitor and communicate DQ scorecard result	Monthly
Monitor	Process 6: DQ Scorecard Monitoring	5	Overall DQ Score	Actual overall DQ Score should be consistently above 95%	>= 95%
-	(To be included in Periodical Review of DQ standards)	6	DQ Standards Periodic Review Frequency	Frequency of reviewing DQ processes to ensure they are up-to-date	Yearly

* To be revised in Operationalization phase based on the experience in pilot

Appendix D: Data Quality Related Forums

DATA GOVERNANCE FORUMS –
DATA QUALITY RELATED FORUMS

A small thumbnail version of the table below, showing the same structure with columns for Forum, Members, Objectives, and Frequency. It includes the same two rows: Data Governance Office (DGO) and Data Quality Working Group.

Forum	Members	Objectives	Frequency*
Data Governance Office (DGO)	- Chairperson(s): DataX/ Monoline DGO - DataX/ Monoline Data Owners & Stewards	- Dedicated resources that champion enterprise data governance and data initiatives - Develop and maintain Data Governance policies, procedures & standards - Escalate unresolved data issues and risk to Data Steering Committee	Monthly
Data Quality Working Group	- Chairperson(s): DataX/ Monoline Data Quality Officer - DataX/ Monoline Data Owners - DataX/ Monoline Business & Technical Data Stewards	- Evaluate and review overall data quality index outcome, focusing on key dimensions identified - Identify data quality improvement actions (i.e. preventive and corrective) and appoint data quality data owner / data steward to lead the activity - Jointly define fix target and expected outcome Manage and track data quality remediation plans based on improvement actions identified	Monthly

*Frequency can be subject to change (e.g. during operationalization, upon completion of setup phase), and forums can be scheduled on ad-hoc basis as needed

END OF DOCUMENT